Redeemability, Digital Currency Adoption, and Flows*

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Abstract

Many digital currencies are backed by redemption promises, but little evidence or theory informs currency redemption policy. We study the impact of redemption policy on digital currency adoption and flows, leveraging novel geographical variation in redemption convenience and transaction data from a peer-to-peer used goods trading platform. We show in a search-theoretic framework that redeemability may induce digital currency adoption at zero steady-state cost. However, the evidence is more consistent with a heterogeneous-agent model, in which tokens flowed on average towards users who were closer to redemption opportunities and redeemability is both costly and necessary for digital money circulation. A key implication is that optimal policy may impose some degree of redemption friction.

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1 Introduction

Many digital currencies are backed by promises of redemption—including emergent stablecoins and platform currencies. However, little research informs the design of currency redemption policy—i.e., choices made by currency issuers that determine the difficulty of redemption. Recent studies characterize optimal adoption subsidies (Alvarez et al. 2023; Crouzet, Gupta, and Mezzanotti 2023) and optimal issuance (Rogoff and You 2023; Rogoff, He, and You 2024). Growing work assesses how redemption policy affects run risk (Huang and Keister 2025; Ma, Zeng, and Zhang 2025). Yet evidence and theory on the steady-state impacts of redemption policy are rare.

The importance of redeemability for the function of money as a medium of exchange is starkly illustrated by historical episodes—from the booms and busts of 18th–19th century banknotes (Volta 1893; Hamilton 1946; Gorton 1996; Velde 2007; Friedman and Schwartz 2008; Sanches 2016) to the recent collapse of the stablecoin UST (Liu, Makarov, and Schoar 2023).[1] Moreover, large corporations such as Walmart and Amazon are considering issuing redeemable digital currencies under the guidance of the 2025 GENIUS Act.[2] To successfully launch and operate redeemable currencies, many questions must be answered. Does redeption opportunities help to encourage currency adoption? How does redemption affect currency flows? Should regulator allow digital currency issuers to restrict redemption? Granular evidence of the impacts of redemption policy on currency operation is thus needed.[3]

In this paper, we study how redemption policy affects steady-state digital currency acceptance, flows, and the profitability of currency issuers. Empirical progress is made by leveraging a unique transaction dataset with rare cross-sectional and time-series variation in

[1] Anthropologists and historians contend that redemption promises, whether by states or private credit institutions, were instrumental in money's earliest emergence (Mitchell-Innes 1913, 1914; Knapp 1924; Lerner 1947; Humphrey 1985; Wray 2004; Graeber 2011). In modern times, regulators employ tools like reserve requirements, deposit insurance, and surveillance to enforce redemption guarantees.

[2] The Guiding and Establishing National Innovation for U.S. Stablecoins Act (GENIUS Act), is a United States federal law that creates a comprehensive regulatory framework for stablecoins. Stablecoins are a type of cryptocurrency that are backed by reliable assets such as a national currency.

[3] Appendix Table ?? shows the detailed examples of redemption-based currency.

currency redeemability. Theoretical progress is made by developing simple search-theoretic models of redeemable money to illustrate how and why redemption policy affects currency acceptance, currency flow, stability, trade, and issuer profits. Our results not only shed light on the impacts of currency redemption policy, but also enable a characterization of optimal redemption policy. Together, they provide useful case evidence and intuitions that can guide the design and operation of emergent digital payment systems.

The unique dataset used in this paper comes from an online trading platform named Bunz, through which a large number of Toronto-based users met in person and traded second-hand items, such as clothing, accessories, plants, and furniture. As previously documented by Wong (2025), the Bunz platform operated a redeemable digital currency, but later discontinued redemption due to financing difficulties. The novelty of the empirical work here is to leverage new cross-sectional variation in redemption convenience to quantify the empirical relationship between token redeemability, acceptance, and flows.

We first document that geographic proximity to redemption opportunities is strongly associated with higher token adoption, net inflows, and redemption, but only mildly or insignificantly higher issuance or holdings in the Bunz economy. Tokens therefore on average flowed from users who are far away from redemption opportunities to those who are closer. This finding reveals that the choice by users to accept tokens was not purely driven by strategic complementary in token acceptance—as is the case in many canonical monetary models—it was influenced in part by their proximity to redemption opportunities. We confirm the robustness of this finding using alternative definitions of redemption convenience and a rich set of individual-level control variables.

We then document that the halt in redeemability caused massive drops in token acceptance and flows, even in areas where redemption was comparatively inconvenient. These reductions are orders of magnitude larger than the effects of a one standard deviation increase in the cross-sectional exposure to redemption prior to the halt. Moreover, initial cross-sectional differences in money acceptance and inflows disappeared. This finding suggests that while strategic complementarities in currency acceptance were significant in this real-world setting, they were not sufficient to sustain a monetary equilibrium in the absence of currency redemption.

While these empirical findings are inherently interesting, it is important to clarify the underlying economic mechanisms. To our knowledge, existing models of money as a medium of exchange do not allow for endogenous redemption. To explain the evidence, we incorporate redeemability into the Kiyotaki-Wright (1993) model of money as a medium of exchange. The novelty is to allow agents to endogenously choose whether to redeem money for a redemption good that yields a utility ν_R upon receiving money.

We first develop a model of redeemable money with homogeneous agents. This benchmark model reveals that currency redemption can be a very low cost method for encouraging currency adoption. Specifically, redeemability can coordinate agents on a monetary equilibrium despite *zero* steady-state redemption. The reason is as follows. An increase in the value of ν_R encourages agents to accept money. Increased money acceptance, in turn, raises the rate at which agents who accept money find profitable trades, so holding money becomes more attractive than redeeming it. Therefore, if the probability of single coincidence is sufficiently high and ν_R is sufficiently large but not too large, there exists a unique monetary equilibrium without any actual token redemption. However, a redemption-run equilibrium—in which agents redeem until the money stock is depleted—also emerges if either ν_R is too low or too high. These findings illuminate the benefits and risk of redeemable money, helping to explain the central role of redeemability in the rise and fall of currencies throughout history. They also explain why currency redemption volume is often low, even though redeemability may be critical for currency adoption.

The homogeneous-agent model does not explain the cross-sectional evidence from Bunz, however. To better match the empirical patterns, we develop a model of redeemable money with agent heterogeneity and partial acceptability. The model reveals that redeemability can become both costly and necessary for sustaining a monetary equilibrium. We focus on environments where the probability of single coincidence is not high, and redemption utility ν_R varies across agents. We find in this environment that money acceptance, in-flows, and redemption all grow with redemption utility ν_R, so money flows from agents with low ν_R to agents with high ν_R. Moreover, redeemability can be necessary for a monetary equilibrium to exist. Our empirical findings are consistent with the heterogeneous agent model, since we find that (1) currency acceptance and flows increase with redemption ease, and (2) currency acceptance dramatically declined when redemption was halt.

As a final step, we provide numerical simulations to characterize optimal redemption policy in the heterogeneous-agent model. We show how agent choices and currency issuer profits vary with the share of agents who receive positive redemption utility. A key finding is that optimal policy may restrict redemption to a limited set of agents, and keep redemption utility only high enough that this set of agent is willing to accept the token, and no higher. This way, the platform minimizes the level of redemption volume, while keeping it high enough to maintain currency circulation. This result helps explain why currency issuers often impose redemption frictions, such as restricting redemption to a subset of agents or charging a redemption fee.

The first and main contribution of this paper is to provide empirical evidence on the impacts of currency redemption policy using novel data and empirical strategies. Our work builds on Wong (2025), who introduced the high-frequency and comprehensive transaction-level data that is used in this paper, and provided time-series evidence on the impact of monetary expansion and reduced redeemability in the Bunz economy. The key innovation in this paper is to study the impacts of currency redemption on currency and trade flows using *cross-sectional* variation in redeemability. We establish new empirical findings, interpret using a novel heterogeneous-agent model with endogenous redemption choices, and empirically highlight the previously underappreciated role of agent heterogeneity and partial acceptability in explaining equilibrium behavior. To our knowledge, this paper is the first to use cross-sectional variation and transaction-level data to empirically explore how currency redemption policy impacts currency acceptance and steady-state flows.

The secondary contribution of this paper is theoretical. Our conceptual framework builds on a variants of Kiyotaki and Wright (1993) that consider how platform or government policy affects the existence and uniqueness of monetary equilibria. Two early papers—Aiyagari and Wallace (1997) and Li and Wright (1998)—show that enforcement of a legal mandate to accept tokens among a subset of the population can engender a unique monetary equilibrium (see also Soller Curtis and Waller 2000; Lotz and Rocheteau 2002; Lotz 2004). Selgin (2003) shows that adaptive learning precludes agents from coordinating on a monetary equilibrium. Wong (2025) provides an homogeneous-agent model where the agents' rate of redemption is *exogenous*, showing that redeemability can help agents coordinate on a monetary equilibrium. Although more recent literature incorporates agent heterogeneity into the popular Lagos-Wright model (e.g., Rocheteau, Weill, and Wong 2021), they do not feature partial acceptability, which is needed to fit our evidence. Our framework builds on Shevchenko and Wright (2004), a first-generation model with agent heterogeneity and partial acceptability.[4] The novelty of our framework is to allow agents to *endogenously* make adoption and redemption choices. In the homogeneous-agent version, we show how a credible promise of redemption can coordinate agents on a unique monetary equilibrium at *zero steady-state cost*, but it also introduces the possibility of a run equilibrium. In the heterogeneous-agent version, we derive testable predictions that are confirmed in data. We also show that redemption frictions are optimal, since they reduce the cost of steady-state currency outflow, while ensuring that the currency successfully circulates.

In a closely related paper, Ma, Zeng, and Zhang (2025) study stablecoin market structure through the lens of a variant of Diamond and Dybvig (1983). They similarly argue that redemption restrictions may be desirable, but for a different reason: they find that redemption restrictions reduce run risk. Huang and Keister (2025) study redemption restrictions for

[4] A related strand of the search-theoretic literature also considers models with bank-issued currencies, but also do not explicitly analyze partial acceptability (e.g. Berentsen, Camera, and Waller 2007; Chiu et al. 2023; Gu et al. 2023; Williamson 2024).

money market mutual funds, arguing that restrictions leave funds susceptible to preemptive runs where investors rush to redeem before the fee applies. Rogoff and You (2023) and Rogoff, He, and You (2024) study optimal issuance policy for redeemable platform currencies. Another large body of work studies the pricing of digital currencies, but do not consider the role of currency redemption policy (e.g., Cong, Li, and Wang 2022; Sockin and Xiong 2023; Garratt and van Oordt 2024).

Our results are especially interesting in light of several recent studies that examine optimal subsidies for encouraging currency adoption using technology adoption models. For example, Crouzet, Gupta, and Mezzanotti (2023) study the 2016 Indian demonetization intervention that increased the relative benefit of firms using electronic payment methods instead of cash. Alvarez et al. (2023) show using a technology diffusion model calibrated to transaction-level data on Costa Rica's national electronic payment system "SINPE" to characterize the optimal subsidy. Alvarez, Argente, and Van Patten (2023) documents the failure of Bitcoin's rollout in El Salvador, despite Bitcoin being declared legal tender. A key implication of such models is that a temporary subsidy can led to a persistent and irreversible increase in adoption. Our approach highlights a different set of policy implications: redeemability can encourage currency circulation in a highly cost-effective manner, but reductions in redemption can dramatically harm currency system operation.

The rest of the paper is organized as follows. Section 2 provides background on the Bunz economy. Section 3 presents our conceptual framework and generates qualitative empirical predictions. Section 4 documents the cross-sectional relationship between redemption convenience and token use. Section 5 reports the heterogeneous impact of reduced redemption. Section 6 characterizes optimal redemption policy. Section 7 concludes.

2 Empirical Setting: Bunz Platform

The Bunz community was founded in 2013 and consisted primarily of young millennial adults in Toronto who arranged to trade second-hand items such as clothing, accessories, plants, and groceries through a mobile app platform. The community's founder forbade cash transactions, so the platform's roughly ten thousand daily active users, who were largely strangers meeting bilaterally in a decentralized manner, initially had to barter.[5]

In April 2018, the Bunz platform introduced a redeemable digital token, BTZ. Each user was endowed with 1000 BTZ upon digital wallet activation. Users could then send BTZ to other users and earn BTZ from the app by answering a survey, inviting friends to join the app, or posting new items. To promote the token, Bunz operated a token redemption program, which allowed users to purchase goods using BTZ at partner local stores at a fixed exchange rate of 100 BTZ to 1 Canadian dollar (CAD).[6] After accepting BTZ payments, the owners of local stores would then receive cash from Bunz HQ at the same fixed exchange rate. The platform did not buy or sell tokens apart from direct issuance to users and redemption at partner stores.

Bunz users can receive BTZ in two ways. First, users receive BTZ from the Bunz Platform by registration, completing specific tasks.[7] Second, users can sell items to other users for BTZ tokens. As for the token outflow, users can either send BTZ to other users to buy their products or redeem the BTZ tokens at merchants cooperating with the Bunz platform to get goods such as coffee, beer, daily necessities, etc. The Bunz Platform will then send these merchants CAD to buy back the BTZ that the merchants hold. Figure A1 illustrates how the token moves between users, merchants, and the Bunz platform.

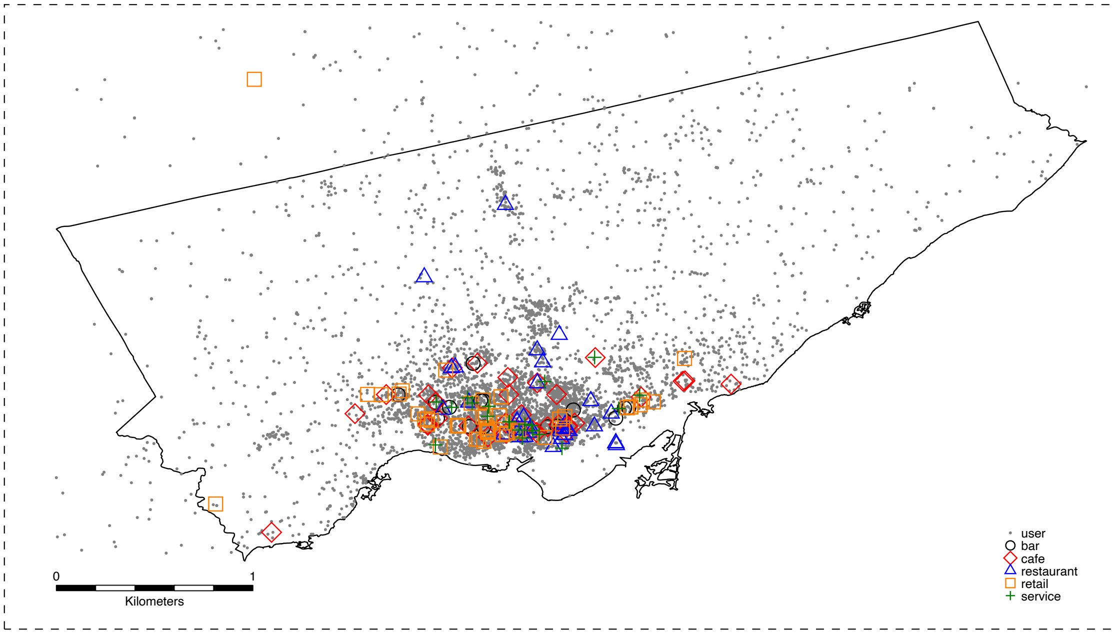
Bunz provided timestamped data for the universe of items posted, messages sent, BTZ transactions, and user ratings after transactions. A unique feature of the data provided by the Bunz platform is that user activity with and without BTZ are both observed at high frequency. The geolocation of a large subset of users is also known. For these reasons, we can

[5] The platform enforced this ban on cash by removing any items asking for cash from the mobile app. Further details are provided in Wong (2025).

[6] In 2018, the average exchange rate was 1 CAD to 0.77 USD.

[7] Bunz may offer BTZ token for watching a video or advertisement, answering questions provided by Bunz, visiting the webpage of a third party such as a Bunz sponsor, and remaining there for a specified time, or other actions or circumstances Bunz may designate from time to time. See https://bunz.com/terms

Figure 1. Map of users and redemption store locations



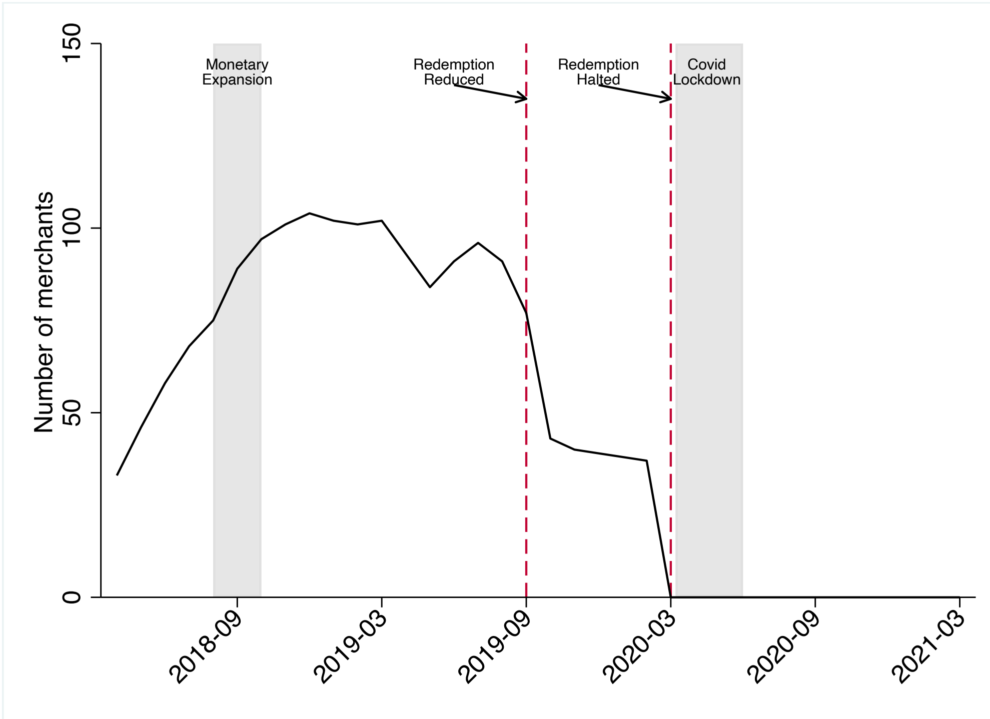
Notes: The map presents the location of the frequent users and redemption stores. Frequent users are defined as users with 20 item posts from April 2018 to August 2019. The users and redemption stores are in an area with a longitude between 79.11524° W and 79.63926° W and a latitude between 43.58100° N and 43.85546° N. This area includes Toronto and parts of its neighborhood.

study how the adoption of digital money depends on a given user's proximity to redemption opportunities.

Figure 1 shows the location of the merchants and active users located in Toronto. As shown in the gray dots on the map, most active users live in the city center of Toronto. Other users live sporadically in Toronto. As for the merchants, in total, 216 merchants at some point accepted BTZ as the payment method, and 155 of the merchants were located in Toronto. These merchants included 50 retail shops, 34 restaurants, 33 cafes, 20 service merchants, 15 bars, 2 beauty merchants, and 1 gallery in Toronto. Most of the merchants in Toronto also locate in the city center of Toronto. Generally, users located in the city center have higher redemption network exposure than user located in other areas.[8] Section 4 studies

[8]Between September and November 2018, Bunz dramatically increased the supply of BTZ through helicopter drops to users in an attempt to drive user traffic. As documented by Wong (2025), the monetary expansion caused large and persistent increases in transaction volume and items posted on the platform among existing users, but did not detectably alter token acceptance patterns.

Figure 2. Total number of token-redeeming merchants over time



Notes: Figure plots the number of active merchants that accept BTZ payments over time. Merchants are defined as active at the month of opening. If merchants no longer have redemption transactions after a specific month, these merchants will be excluded from active merchants.

the cross-sectional relationship between token use and redemption convenience during the period when the BTZ redemption program was in operation.

Figure 2 shows the number of active merchants over time. We define merchants as active when they start to accept BTZ payments. Merchants do not accept BTZ redemption after a specific month. In that case excluded after that month. From April 2018 to December 2018, the number of active merchants increased continuously. Bunz also expanded the monetary supply from August 2018 to October 2018, after which the introduction of merchants stabilized. On September 10, 2019, Bunz halted on redemption at retail and service-providing stores without giving any prior notice, causing some users to stop accepting BTZ and rush to the remaining merchants to redeem their BTZ. On February 26, 2020, Bunz completely halted the Shop Local program but called it a temporary pause. In prior work, Wong (2025) showed that the BTZ price of posted items remained largely stable throughout the observable period.

3 Conceptual Framework

In this section, we construct two deliberately simple models to generate qualitative predictions mapping to our empirical results. Both feature the endogenous decisions to accept and redeem money. We begin with a tractable homogeneous-agent benchmark to build intuition. Subsequently, we introduce agent heterogeneity to develop a more realistic model that accounts for the key empirical findings.

3.1 Benchmark: Model with Homogeneous Agents

In our homogeneous-agent model, money must solve a coordination problem in order to facilitate trade. It is profitable for agents to accept money only if they expect others to do the same, which gives rise to two possible outcomes: a monetary equilibrium in which money circulates and a non-monetary equilibrium in which it does not. We show that redemption opportunities can help resolve this coordination problem and guarantee that money circulates. Our analysis yields two additional insights. First, money circulation can be sustained at very low cost, since equilibrium redemption volume may be zero in steady state. Second, depending on redemption terms and agents' strategic beliefs, the economy may also feature a run equilibrium in which agents rush to redeem.

The model features agents that want to trade but need to overcome search frictions arising from the lack of coincidence of wants. The set of agents in the economy is denoted by N, with measure $\mu(N) = 1$. Each agent i can produce a unit of a certain type of goods, G^i, and can consume only one type of goods g^i. Agents cannot consume their own products, so they have to meet and exchange goods with other agents in order to consume. Goods are perishable, and production is instantaneous. Each agent derives utility $u_C > 0$ from consuming a good, incurs cost $c > 0$ from producing a good, and we assume that $u_C - c > 0$. Each agent discounts future utility with time preference $\beta > 0$.

Time is discrete. In each period, agents meet with probability $\alpha > 0$. The probability that agent i meets another agent whose product i can consume is $P(g^i \in G^j) = x$. For simplicity, we assume that the probability of a "double coincidence of needs" is $P(g^j \in G^i \mid g^i \in G^j) = 0$. We let $l = \alpha x$ denote the probability of a "single coincidence" meeting.

In this environment, money facilitates trade by allowing agents to transact even in the absence of a double coincidence of wants. Money is indivisible, durable, and has zero storage cost. The money supply is denoted by $M > 0$. We assume that one unit of money is always traded for one unit of commodity. This finding matches the fact that there was no observed inflation in the Bunz economy. It can also be rationalized by models with price coordination frictions (e.g., Green and Zhou 1998, 2002; Kamiya and Shimizu 2006, 2007a,b, 2011; Jean, Rabinovich, and Wright 2010).

Agents' decisions. In each meeting, agents make two key decisions: whether to accept money in exchange for goods, π, and whether to redeem money once they hold it, ρ.[9]

When agents meet in pairs, they barter and consume upon a double coincidence of needs. Upon a single coincidence of needs, the agent with the ability to produce the desired good faces a decision problem of whether to accept money from their transaction partner. For simplicity, we assume that agents can hold at most one unit of money, so only agents who do not hold money can choose to accept money. We denote the acceptance choice by $\pi \in \{0, 1\}$. If the agent holds a unit of money, the agent can choose to redeem and enjoy utility flow ν_R, use the money to transact if she meets a money-accepting agent, or store it in anticipation of a future transaction. We denote the redemption choice as $\rho \in \{0, 1\}$.

Thus, agents can transition between two states: state 0 of not having money, or state 1 of receiving one unit of money. We denote the measure of agents in state 1 as μ_1. Then, given agents' decision problem, their value functions are characterized by the following equations.

[9] Since agents are fully symmetric in their production and homogeneous on all other aspects, we suppress the index i on the decision variables.

$$V_t^1 = \max_{\rho} \underbrace{\rho\left[\nu_R + \beta V_{t+1}^0\right]}_{\text{redemption value = direct utility and cont. value}} \tag{1}$$

$$+ (1-\rho)\underbrace{\Big[\underbrace{l\Pi(1-\mu_1)u_C}_{\text{consumption utility from trade}} + \beta\big(\underbrace{l\Pi(1-\mu_1)V_{t+1}^0 + \big(1 - l\Pi(1-\mu_1)\big)V_{t+1}^1}_{\text{cont. value from trade}}\big)\Big]}_{\text{transaction value from holding money as a buyer}}$$

$$V_t^0 = \max_{\pi} \underbrace{\underbrace{-lM\pi c}_{\text{production cost}} + \beta\big[\underbrace{lM\pi V_{t+1}^1 + (1-lM\pi)V_{t+1}^0}_{\text{cont. value from trade}}\big]}_{\text{transaction value from selling goods}} \tag{2}$$

Solution concept. We focus on steady-state symmetric pure-strategy Nash equilibria. From the agents' point of view, there are two uses of money: redemption and transaction. The value of these two uses depend on ν_R and u_C. Depending on the parameters, a multiplicity of equilibria may arise.

An equilibrium is labeled "non-monetary" if agents neither accept nor redeem money ($\pi = 0, \rho = 0$). An equilibrium is "monetary" if agents accept money and do not redeem ($\pi = 1, \rho = 0$). We define an equilibrium as a "run" if agents redeem ($\rho = 1$). Since we assume there is no issuance, the steady-state money stock in run equilibria is zero ($M = 0$). There are two types of run equilibria: one where agents accept ($\pi = 1$), and one where agents do not ($\pi = 0$). To characterize steady-state equilibria, we define $\underline{u} \equiv \frac{c(1-\beta+\beta l(1-M))}{\beta l(1-M)}$ and $\bar{\nu}(x) \equiv \frac{l(1-x)u_C + \beta lxc + \beta l^2 x(1-x)(u_C - c)}{1-\beta+\beta l}$. Note that $\bar{\nu} > 0$ and $\bar{\nu}' < 0$.

Proposition 1. *Suppose $u_C \geq \underline{u}$. Then:*

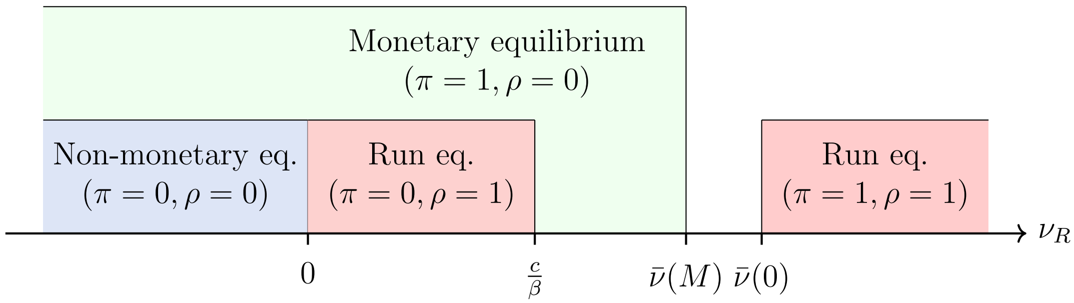
1. *A non-monetary equilibrium with $M > 0$ exists if and only if $\nu_R \leq 0$;*

2. *A monetary equilibrium with $M > 0$ exists if and only if $\nu_R \leq \bar{\nu}(M)$; and*

3. *A run equilibrium with $M = 0$ exists if and only if $\nu_R \in [0, c/\beta]$ or $\nu_R \geq \max\{\frac{c}{\beta}, \bar{\nu}(0)\}$.*

Proof. See Appendix. $\square$

Figure 3. Steady-state equilibria as a function of ν_R



Proposition 1 states that a non-monetary equilibrium exists only when ν_R is nonpositive. When ν_R is positive, agents either accept money or redeem it. If c is sufficiently low and ν_R is sufficiently large but not too large, there exists only monetary equilibria. This occurs since agents who accept money find profitable trades, so holding money becomes more attractive than redeeming it. However, a redemption-run equilibrium—in which agents redeem until the money stock is depleted—also emerges if either ν_R is too low or too high. Figure 3 illustrates this result. Corollary 1 states the conditions under which redeemability can coordinate agents on a monetary equilibrium despite *zero* steady-state redemption.

Corollary 1. *Suppose $u_C > c(1 - \beta(1 - l))/l\beta$. Then there exist $\nu_R > 0$ such that at least one steady-state equilibrium exists and all equilibria are monetary.*

3.2 Model with Heterogeneous Agents

To account for the richness of our setting, this subsection modifies the model to incorporate agent heterogeneity. We build on Shevchenko and Wright (2004), which, to our knowledge, is the only readily available model for analyzing economies with agent heterogeneity and partial acceptability. Two results are derived. First, agents with more redemption opportunities are more likely to accept money, leading to a flow of money from agents with few redemption opportunities to those with more redemption opportunities. Second, a positive steady-state redemption volume may be necessary for a monetary equilibrium to exist.

The model features two forms of agent heterogeneity. First, agents are heterogeneous in terms of the utility they derive from consuming transacted goods. We denote agent i's consumption utility as u_C^i. For simplicity, we assume that u_C^i is distributed uniformly on a support $[\underline{u}_C, \overline{u}_C]$ across the distribution, where $\underline{u}_C > 0$. Second, the issuer offers heterogeneous redemption utilities to agents, denoted as $\nu_R^i \geq 0$.

To keep the steady-state money supply constant in the economy, we assume that money is exogenously issued to agents in each period. Specifically, agents who hold no money receive a unit of money with probability σ. This assumption matches the way the Bunz platform issues money to users via helicopter drops in our empirical setting.

Under these assumptions, the enriched Bellman equations of the agent i are given by

$$V_{1,t}^i = \underbrace{\rho_t^i \left[\nu_R^i + \beta V_{0,t+1}^i\right]}_{\text{redemption value}} + \underbrace{(1-\rho_t^i)\Big[lW_t u_C^i + \beta\big(lW_t V_{0,t+1}^i + \big(1-W_t\big)V_{1,t+1}^i\big)\Big]}_{\text{transaction value from holding money as buyer}} \tag{3}$$

$$V_{0,t}^i = \max_{\pi_t^i} \underbrace{-lM_t\pi_t^i c + \beta\left[lM_t\pi_t^i V_{1,t+1}^i + (1-lM_t\pi_t^i)V_{0,t+1}^i\right]}_{\text{transaction value from selling goods}} + \underbrace{\beta\sigma(V_{1,t+1}^i - V_{0,t+1}^i)}_{\text{issuance value}}. \tag{4}$$

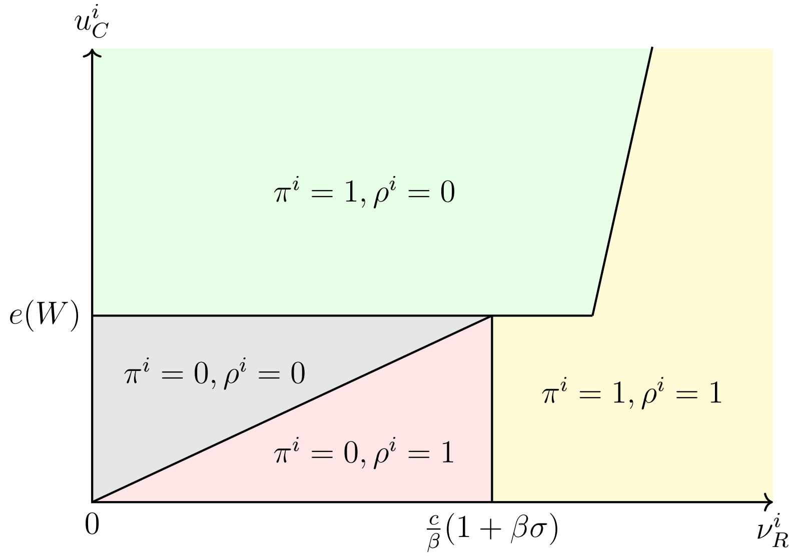
Following the previous section, π_t^i denotes agent i's probability of accepting money in a single-coincidence meeting and ρ_t^i denotes her probability of redeeming money upon receiving it. In addition, μ_t^i denotes the probability that agent i is in state 1 in period t. $M_t = \sum_j \mu_{t-1}^j(1-\rho_{t-1}^j)$ denotes the money stock, while $W_t = \sum_j \pi_{t-1}^j(1-\mu_{t-1}^j)$ is the aggregate probability that agent i meets an agent that accepts money.

As before, we define a steady state equilibrium where aggregate quantities and individual optimal decisions, $(M, W, \{\mu^i\}_i, \{\pi^i\}_i, \{\rho^i\}_i)$, are constant in time.

We first study agents' optimal individual decisions as a function of their consumption utility u_C^i and redemption utility ν_R^i. We relegate the details of analyses to the Appendix and directly present the structure of optimal decisions in the following Lemma.

Lemma 1. *Given $W, M \in (0,1)$, each agent i's optimal choice (π^i, ρ^i) as a function of (u_C^i, ν_R^i) is given by Figure 4.*

Figure 4. Optimal individual acceptance and redemption in heterogenous-agent model



Notes. This figure presents how each agent i's optimal acceptance decision π^i and redemption decision ρ^i depend on u_C^i, ν_R^i. Cutoffs between regions are functions of the aggregate states and primitives.

Proof. See Appendix. □

Figure 4 shows that agents accept money if and only if either (a) redemption utility ν_R^i exceeds a certain cutoff $\frac{c}{\beta}(1+\beta\sigma)$, or (b) consumption utility u_C^i is large relative to a cutoff $e(W) = \frac{c}{\beta}\frac{1+\beta\sigma-\beta(1-lW)}{lW}$, which decreases in lW, the aggregate probability that agent i meets an agent that accepts money. Among these agents, those whose u_C^i is large relative to ν_R^i do not redeem. Agents with both low u_C^i and low ν_R^i do not accept money. However, some may redeem whatever money that they are issued, if ν_R^i high when compared to u_C^i .

The optimal acceptance and redemption choice characterized above imply that money will on average flow from agents who do not redeem towards agents who redeem in steady-state equilibrium. Moreover, as money on average flow towards the redeeming agents, goods accordingly on average flow away from redeeming agents.

To see this formally, let money inflows from peers to agent i be $S^i = lM(1-\mu^i)\pi^i$, i.e., the expected number of transactions in which agent i accepts money in exchange for a produced good. Let money outflows from peers be $P^i = lW\mu^i(1-\rho^i)$, i.e., the expected number of transactions in which agent i uses money to obtain a good.

Proposition 2. *In any steady-state equilibrium, consider any two agents i, j with consumption utility $u_C^i = u_C^j$ and redemption utility $\nu_R^i < \nu_R^j$. Their behaviors satisfy:*

1. *The probability of accepting money increases with redemption utility ($\pi^i \leq \pi^j$).*

2. *The volume of money inflows from peers increases with redemption utility ($S^i \leq S^j$).*

3. *The volume of money outflows from peers decreases with redemption utility ($P^i \geq P^j$).*

Proof. See Appendix. □

We next examine what types of equilibria exist in the heterogeneous agent model. Here we take the issuer's redemption policy as exogenous. We extend the proof strategy developed in Shevchenko and Wright (2004) to characterize two dimensions of equilibrium choices among a large and heterogeneous population. Here we say that an equilibrium is *non-monetary* if all agents play $\left(\pi^i = 0, \rho^i = 1\right)$. We say that an equilibrium is *monetary* if all agents play either $\left(\pi^i = 1, \rho^i = 0\right)$ or $\left(\pi^i = 1, \rho^i = 1\right)$.

The following Proposition states that when agents are sufficiently heterogeneous and redemption utilities cannot be targeted, the economy converges to a unique *non-monetary* equilibrium if redemption utilities for all agents are below a cutoff value. If instead redemption utilities for all agents are above the same cutoff value, it converges to a unique monetary equilibrium.

Proposition 3. *Suppose the dispersion of u_C^i among agents is large*[10] *and that u_C^i and ν_R^i are independently distributed. Then there exists $\overline{\nu} > \frac{c}{\beta} + \sigma$ such that:*

1. *If $\nu_R^i < \frac{c}{\beta} + \sigma$ for all agents i, then there exists a unique* non-monetary *equilibrium.*

2. *If $\nu_R^i \in [\frac{c}{\beta} + \sigma, \overline{\nu}]$ for all agents i, then there exists a unique* monetary *equilibrium.*

[10]Formally, see Assumption 1 in the Appendix.

Proof. See Appendix. □

Proposition 3 states that the unique equilibrium is non-monetary when redemption utilities are zero. This result contrasts with Proposition 1, which showed that there exists both a monetary equilibrium and a non-monetary equilibrium when redemption utility is zero. Assumption 1 is critical for this difference. Since the dispersion in u_C^i is large, the presence of agents who actively redeem from the issuer is now *necessary* for the existence of the monetary equilibrium. In other words, steady-state redemption volume is now necessary for sustaining money circulation.

3.3 Testable Predictions

The heterogeneous-agent model has two empirical predictions that are not available in the homogeneous-agent model. These two predictions will be tested in the following sections.

Prediction 1. *In any monetary equilibrium, money acceptance and in-flows increases with redemption utility v_R in the cross-section.*

Section 4 tests this prediction by exploiting cross-sectional variation in users' access to redemption opportunities on the Bunz platform. We proxy an individual user's redemption utility by their geographic proximity to token-redeeming merchants, measured as the number of merchants within walking distance of the user. We then examine whether users with greater redemption exposure are more willing to accept the token in peer-to-peer trades and whether they receive more tokens from other users. By comparing token acceptance rates and token inflows across users who are otherwise similar but differ in their proximity to redemption outlets, we assess whether money acceptance and inflows systematically increase with redemption convenience, as implied by the heterogeneous-agent model.

Prediction 2. *If there is sufficient dispersion in consumption utility u_C, then redeemability may be necessary for money circulation. If redemption is halted, there is a transition from a monetary equilibrium to a non-monetary equilibrium.*

Sections 5 and 6 test this prediction by studying how token acceptance and circulation respond to sharp, unexpected reductions in redeemability. We leverage two policy changes on the platform, first a partial restriction of redemption and later a complete halt, to examine whether the economy transitions away from a monetary equilibrium when redeemability is reduced or eliminated. We track the evolution of token acceptance, redemption, and flows before and after these events, and compare responses across users with different initial exposure to redemption opportunities. This approach allows us to assess whether the removal of redeemability leads to a large, system-wide collapse in token acceptance and circulation, consistent with redeemability being necessary to sustain monetary exchange when agents are heterogeneous.

4 Effects of Redemption Convenience on Token Usage

Do redemption opportunities matter for digital currency adoption? Prediction 1 suggests that token acceptance and inflows increases with redemption utility ν_R^i in an economy with agent heterogeneity. In this section, we test this proposition by measuring how proximity to redemption opportunities—as measured by the number of nearby redemption merchants—affects user token acceptance, flows, and holdings. We robustly confirm, using various regression specifications, that redemption opportunities increased token acceptance and inflows, and that tokens flowed on average from users farther away from redemption opportunities towards those closer.

4.1 Methodology

To measure user-level exposure to redemption opportunities, we leverage the fact that a large number of users provide their geolocation to the platform, so their physical distance to redemption merchants can be calculated. We focus on the period between April 2018 and August 2019, when the redemption program was at its full extent. We focus on frequent users, defined as those who posted 20 or more items during this period, and exclude users who posted more than 70 percent of their items in a single month. Since geolocation is not available for all users, we focus on users for whom an exact geolocation within Toronto is observed.[11] This leaves 7,162 users. This sample accounts for 53 percent of token redemption, 60 percent of ratings, 47 percent of token inflows, and 57 percent of token outflows (see Appendix Tables B1 and B2).

We define user-level *redemption exposure* as the average number of redemption merchants within 1 km of users from April 2018 to September 2019. This variable captures the number of redemption stores that the users easily access on foot. We then measure how redemption exposure affects user behavior using the following regression specification:

$$y_i = \beta \times Exposure_i + \gamma X_i + \epsilon_i, \tag{5}$$

where y_i represents outcomes including token redemption, acceptance, flows, and holdings of user i. β is the estimated relationship between the number of merchants within 1 km of users and users' behavior related to tokens. X_i is the demographic characteristics, activeness, and the distance to the city center of users. The demographic data includes the age, income, and education level.[12] The measurements of activeness include the number of item posts and the number of completed trades — measured by the number of ratings sent to other users[13]. The main threat to causal identification is the fact that users closer to redemption stores may be selected. Since the estimates are highly similar even when additional controls are added, it is likely that the estimates at least partly capture a causal effect of redemption exposure.

4.2 Main results

Table 1 reports the main result of this section—namely, that redemption exposure increased token acceptance. We measure token acceptance as the proportion of items a user

[11] 32.94 percent of users provide only their city of residence, rather than a specific geolocation within the city.

[12] As the demographic data is the survey data of Bunz, we only have 2,204 users who have completed these surveys.

[13] Upon completion of a transaction, both parties may rate each other. If neither party submits a rating, the system will automatically assign a default rating.

Table 1. Effect of redemption exposure on token acceptance

	(1)	(2)	(3)	(4)	(5)
	Dependent variable: Token acceptance				
# Nearby redemption merchant	0.003*** (0.001)	0.004*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003** (0.001)
Demographic controls		X			X
User activeness controls			X		X
Distance to city center				X	X
# Obs	7,162	2,204	7,162	7,162	2,204

Notes: Table shows the effect of redemption exposure on token acceptance. The nearby redemption merchant is defined as the number of merchants within 1 km of users. The baseline mean is the average token acceptance of users with zero redemption exposure. The analysis period is from April 2018 to August 2019. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

lists with a BTZ-denominated price, which signals the user's willingness to transact in tokens. Column (1) indicates that having one additional merchant within a 1-kilometer radius is associated with a statistically significant 0.3 percentage point (p.p.) increase in token acceptance. Columns (2) through (5) show that this positive relationship is robust to additional control variables.

Table 2 shows that redemption exposure not only affected acceptance, but also the flow of tokens. Panel A Column (1) shows that an additional nearby merchant is associated with a statistically significant 2.6 percent increase in token inflows (i.e., the number of tokens received by the user from other users). The economic magnitude and statistical significance of this effect is similar across Columns (2)-(5), as control variables are added.

In contrast, Panel B shows limited correlation between redemption exposure on token outflows (i.e., the number of tokens sent by the user to other users). Column (1) shows that an additional merchant within 1 km is associated with a statistically insignificant 0.2 percent reduction in token outflows. Columns (2) - (4) shows that the estimated effect remains stable when controlling for user demographics, activity levels, and distance to city center. Column (5) reveals a marginally significant positive effect after adding all controls. Overall, these regression reveal that tokens on averaged flowed from uses farther away from redemption

Table 2. Effect of redemption exposure on token inflows and outflows

	(1)	(2)	(3)	(4)	(5)
	Panel A: Asinh token inflow				
# Nearby redemption merchant	0.026***	0.030***	0.023***	0.026***	0.039***
	(0.005)	(0.008)	(0.003)	(0.006)	(0.007)
	Panel B: Asinh token outflow				
# Nearby redemption merchant	0.002	-0.007	-0.002	0.003	0.011*
	(0.004)	(0.008)	(0.003)	(0.006)	(0.006)
Demographic controls		X			X
User activeness controls			X		X
Distance to city center				X	X
# Obs	7,162	2,204	7,162	7,162	2,204

Notes: Table shows the effect of redemption exposure on asinh ($asinh(x) = ln(x + \sqrt{x^2+1})$) amount of token inflow and outflow. The nearby redemption merchant is defined as the number of merchants within 1 km of users. The baseline mean is the average token inflows and outflows of users with zero redemption exposure. The analysis period is from April 2018 to August 2019. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

merchants to those closer.

Table 3 examines how redemption exposure affected token redemption (i.e., the number of tokens accepted by the platform from the user) and token issued by the Bunz platform to the user (i.e. the number of token transferred from Bunz to users in response to user actions such as filling out surveys on the app).[14] Panel A Column (1) shows that an additional merchant within 1 km is associated with a statistically significant 6.9 percent increase in tokens redeemed. Columns (2) - (5) shows that this positive association is robust to additional controls.

Panel B Column (1) shows an economically small and statistically insignificant effect (0.4 percent increase) of additional merchants on tokens issued. Columns (2) and (3) show that this null result persists when controlling for demographics and user activity. However, Columns (4) and (5) account for distance to city center, reveal statistically significant effects of 1.5 percent and 0.7 percent, respectively, which remain substantially smaller than the

[14]Bunz distributes tokens for various user actions, including watching videos or advertisements, completing surveys, visiting sponsor webpages, or other designated activities. See https://bunz.com/terms for details.

Table 3. Effect of redemption exposure on token redemption and issuance

	(1)	(2)	(3)	(4)	(5)
	Panel A: Asinh token redeemed				
# Nearby redemption merchant	0.029***	0.044***	0.029***	0.023***	0.031***
	(0.003)	(0.007)	(0.003)	(0.004)	(0.008)
	Panel B: Asinh token issuance				
# Nearby redemption merchant	0.004	-0.004	0.002	0.015***	0.007*
	(0.004)	(0.004)	(0.003)	(0.005)	(0.004)
User demographic controls		X			X
User activeness controls			X		X
Distance to city center				X	X
# Obs	7,162	2,204	7,162	7,162	2,204

Notes: Table shows the effect of redemption exposure on asinh amount of token redeemed and issuance. The nearby redemption merchant is defined as the number of merchants within 1 km of users. The baseline mean is the average token redemption and issuance of users with zero redemption exposure. The analysis period is from April 2018 to August 2019. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

effects of redemption exposure on token redeemed.

Table 4 reports the effects of redemption exposure on user token holdings. Column (1) shows that an additional merchant within a 1 km radius of users corresponds to a statistically insignificant 0.3 percent increase in token holdings. Columns (2) and (3) further confirm the lack of significance after controlling for users' demographic characteristics and activeness, respectively. However, Columns (4) and (5) reveal a statistically significant positive relationship: an additional merchant is associated with a 1.4 percent and 1.7 percent increase in token holdings after controlling for distance to the city center.

4.3 Discussion and Additional Analyses

The above estimates reveal that token acceptance, inflows from peers, and redemption strongly increases with proximity to redemption opportunities. Meanwhile, token outflows to peers, issuance, and holdings had weak relationships with redemption exposure. In other words, tokens on average flowed from users who were farther away from redemption opportunities towards those who were close. This finding is consistent with the heterogeneous-agent

Table 4. Effect of redemption exposure on token holdings

	(1)	(2)	(3)	(4)	(5)
	Dependent variable: Asinh token holding				
# Nearby redemption merchant	0.003 (0.004)	0.004 (0.005)	0.001 (0.004)	0.014** (0.006)	0.017*** (0.006)
Demographic controls		X			X
User activeness controls			X		X
Distance to city center				X	X
# Obs	7,162	2,204	7,162	7,162	2,204

Notes: Table shows the effect of redemption exposure on asinh amount of token holdings. The nearby redemption merchant is defined as the number of merchants within 1 km of users. The baseline mean is the average token holdings of users with zero redemption exposure. The analysis period is from April 2018 to August 2019. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

model, wherein acceptance does not purely depend on strategic complementarities between agents, but also on proximity to redemption opportunities.

This subsection discusses additional results that shed further light on the underlying mechanisms. First, we study how redemption exposure affected the intensive vs. extensive margins of token flows——namely, the number of transactions and the probability of transactions. Appendix Tables B3, B4, B5 and B6 report the effects of redemption exposure on these extensive margin measures of token inflows, outflows, redemption, and issuance, respectively. They show that redemption exposure increases the extensive margin measures of token inflows and token redemption, but to a much smaller extent than the change in total inflow and redemption. We also find limited correlation between both extensive and intensive margin responses in both token outflows and issuance. These results suggest that the effects are to a substantial extent due to intensive margin responses.

We next ask: Which types of stores drove the above correlations? Specifically, we examine how token acceptance, flows, and holdings were affected by exposure to food and retail merchants, respectively, by estimating the following specification:

$$y_i = \beta_f \times Exposure_{f,i} + \beta_r \times Exposure_{r,i} + \gamma X_i + \epsilon_i, \tag{6}$$

where $Exposure_{f,i}$ and $Exposure_{r,i}$ denote the number of food and retail merchants within 1 km of user i, respectively.[15] Here, the coefficients β_f and β_r capture the association between the user's token-related behavior and the number of food and retail merchants located within 1 km of the user, respectively. Appendix Table B7 reports bivariate regression estimates. Appendix Table B8 adds controls for users' demographic characteristics, activity levels, and distance to the city center.

The results reveal that token acceptance is more positively correlated with food merchant exposure than with retail merchant exposure. Specifically, one additional food store within 1 km of a user corresponds to a 0.6 p.p. increase in the share of new items priced in tokens, whereas one additional retail merchant increases acceptance by only 0.1 p.p. Similarly, token inflows, outflows, and issuances are more positively correlated with food merchant exposure rather than with retail merchant exposure.

Intriguingly, however, token redemption is more positively correlated with *retail* merchant exposure than with food merchant exposure. One additional food store within 1 km of a user corresponds to a roughly 4 percent increase in redemption, whereas one additional retail merchant increases redemption by roughly 8 percent. These results suggest that retail stores (which sell nonperishable goods) impose a larger redemption burden on the platform, without a significant benefit of increasing token acceptance, when compared to food stores (which only sell perishable items).

Finally, several robustness checks confirm that the estimates are not specific to early users, attributable to differences in user activity profiles, or specific to the chosen redemption exposure measure. Appendix Table B9 re-estimates the effects of redemption exposure using a restricted sample of users who posted at least 20 items prior to token introduction, showing that the results are qualitatively unchanged. Appendix Table B10 estimates Equation (5) with the number of item posts and completed trades as the outcome variables, showing

[15] Food merchants are defined as cafes and restaurants; retail merchants are defined as retail shops, bars, and service shops.

that the number of item posts and completed trades are not correlated with redemption exposure even after controlling for demographic characteristics and distance to city center. Appendix Table B11 estimates Equation (5) using alternative measures for redemption exposure, showing that our main findings regarding token activities highly similar. Appendix Tables B12 and B13 estimate Equation (5) using the inverse hyperbolic sine (asinh) of token flows at different scales (e.g., 1 percent, 100 times and 10,000 times the original flow) as outcome variables; the results remain qualitatively unchanged. Appendix Table B14 further estimates Equation (5) using a Poisson regression model. These estimates indicate that the results for token inflow, redemption, and issuance are qualitatively similar, though the effect of redemption exposure on token outflow becomes significantly positive.

5 The Heterogeneous Effects of Reduced Redeemability

How does a reduction in redeemability differentially affect acceptance and flows for users with different initial exposure to redemption opportunities? Prediction 2 suggests that reduced redeemability triggers a transition from a monetary equilibrium, in which tokens are accepted, to a non-monetary equilibrium, where token acceptance is much lower. As a consequence, changes in redeemability may result in changes in token acceptance that are much larger than initial cross-sectional differences.

To test this prediction, this section estimates the heterogeneous effects of an unexpected event wherein redeemability was sudeently reduced on Bunz platform. We find that reduced redeemability caused a overall reduction in token acceptance, but did not eliminate differences in token acceptance among users with different redemption exposure. There was also a temporary spike in token redemption, as users attempted to spend down their token balances. The spike was substantially larger for users with higher redemption exposure.

5.1 Methodology

On September 9, 2019, Bunz HQ announced that tokens would only be redeemable at local partner stores that sell coffee or food. This policy change was primarily driven by significant cash flow issues the company faced in late 2019. The following day, Bunz HQ updated the broader community through a blog post, confirming that it would no longer accept BTZ outside of coffee shops and restaurants.

Our analysis of the impact of reduced redeemability focuses on a 26-week window surrounding the policy change, spanning 11 weeks before to 15 weeks after. First, we plot the 7-day moving average of the token acceptance and redemption[16] of frequent users with high, low, and zero initial redemption exposure, as measured by average number of merchants within 1 km. Users with zero initial redemption exposure are those without nearby merchants. Users with high (low) initial redemption exposure are those with above (below) median average merchants within 1 km, excluding those with zero redemption exposure.[17]

Next, we quantify the effect of reduced redeemability on token activities and the influence of redemption exposure using the following regression equation.

$$y_{i,t} = \sum_{k=-11}^{15} (\beta_{I,k} I_{t-t^*=k} + \beta_{D,k} Exposure_i \times I_{t-t^*=k}) + \delta_i + \varepsilon_{i,t} \tag{7}$$

where y_{it} denotes asinh outcomes of user i in week t, including token acceptance and redemption[18], $I_{t-t^*=k}$ are dummies indicating each week relative to when redeemability was reduced, $Exposure_i$ is the the number of merchants within 1 km of users, δ_i are user fixed effects, and $\varepsilon_{i,t}$ is an error term. $\beta_{D,k}$ and $\beta_{I,k}$ represent the interaction effect and main effect, respectively. Standard errors are clustered at user level.

5.2 Main Results

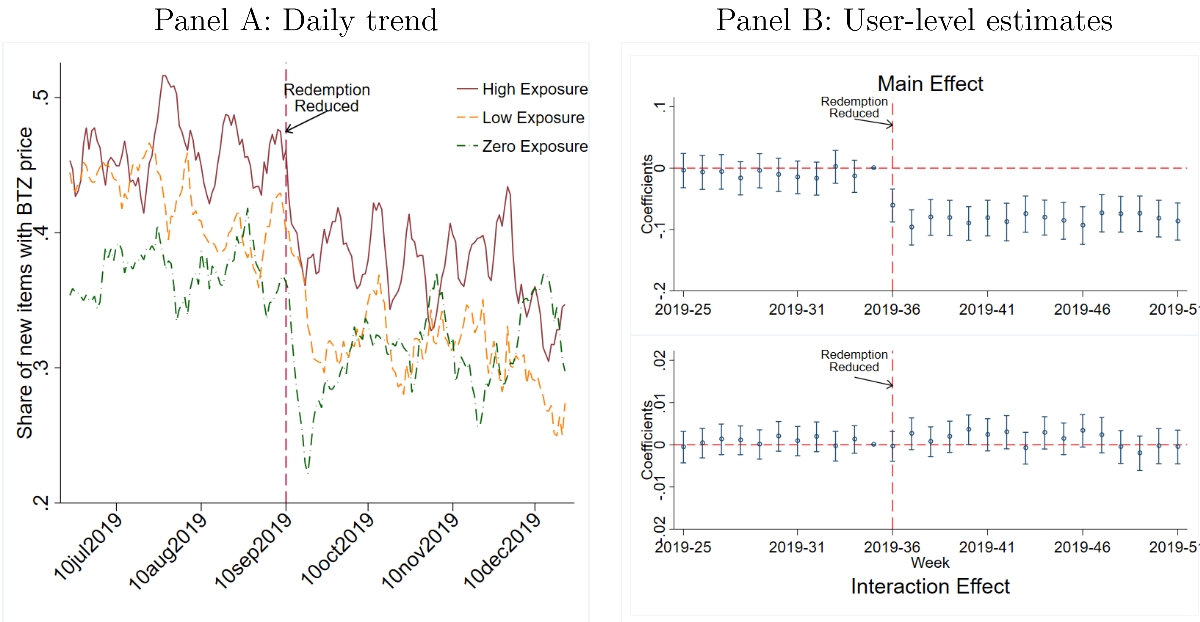
Figure 5 reports the main result of this section—namely, the impact of reduced redeemability on token acceptance, as measured by the share of items with posted BTZ prices. Panel

[16] In Appendix C, we further show the trend of token inflow, outflow, issuance, and holdings.

[17] The distribution of redemption exposure is shown in Appendix Figure C1: The majority of users do not have any redemption stores within 1km, but the extent of redemption exposure is quite widely dispersed otherwise.

[18] We also use the token inflow, outflow, issuance, and holdings as outcome variables and plot the coefficients in Appendix C.

Figure 5. Effects of reduced redeemability on token acceptance

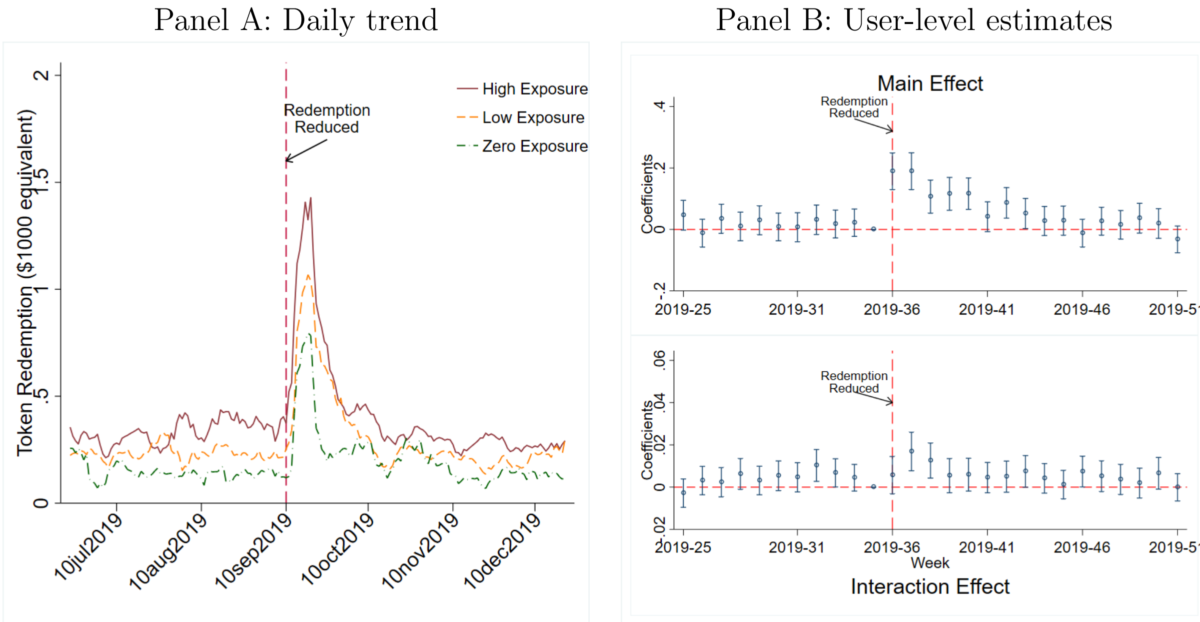


Notes: The figure plots the effect of reduced redeemability on token acceptance. Panel A shows the 7-day moving average share of new items with a posted BTZ price by redemption exposure. Panel B plot the regression coefficients $\beta_{I,k}$ and $\beta_{D,k}$, with the 95 percent confidence intervals, estimated by Equation (7) using the share of new items with a posted BTZ price as outcome variable. The red dashed line indicates September 10, the day of partial cessation of Shop Local program.

A show raw averages among group of users sorted by redemption exposure, revealing that token acceptance immediately dropped by approximately 8 p.p. in all groups. Furthermore, users with higher redemption exposure still have higher token acceptance level than users with lower redemption exposure. Panel B reports regression coefficients, with controls for user fixed effects. It reveals that token acceptance immediately fell by 6-9 p.p. among existing regular users, conditional on their posting new items. Moreover, the correlation redemption exposure and token acceptance does not significantly change after the reduction in redeemability.

Figure 6 documents the impact of reduced redeemability on realized token redemptions. Panel A shows that users on average redeemed more tokens after the partial halt of redemption program and fell to the initial level afterwards. Users with higher redemption exposure

Figure 6. Effects of reduced redeemability on token redemption



Notes: The figure plots the effect of reduced redeemability on token redemption. Panel A shows the 7-day moving average amount of token redeemed by redemption exposure. Panel B plot the regression coefficients $\beta_{I,k}$ and $\beta_{D,k}$, with the 95 percent confidence intervals, estimated by Equation (7) using the token redeemed as outcome variable. The red dashed line indicates September 10, the day of partial cessation of Shop Local program.

also redeem more tokens during our analysis period. Panel B reports that token redemption increased on average by roughly 15 percent in the first four weeks and fell to the level before redemption reduction in the long run. Additionally, users with one more merchants redeemed further 2 percent mroe tokens in the following two weeks and drops afterwards.

5.3 Discussion and Additional Results

The above estimates confirm that reduced redeemability led to a decline in token acceptance and a transient spike in redemption activity across the platform, as previously shown by Wong (2025). In addition, they reveal that reduced redeemability did not eliminate the acceptance disparities among users with varying levels of redemption exposure, and led to a larger temporary redemption spike among users closer to redemption opportunities. These findings confirm that changes in redeemability may result in changes in token acceptance much larger than initial cross-sectional differences, and may trigger redemption-run responses that are heterogeneous according to redemption exposure.

Several additional results enrich the above analysis.

First, we examine that the heterogeneous impacts on reduced redeemability on other outcome variables. The findings are highly consistent with the conclusions above.

Appendix Figures C2 and C3 examine the heterogeneous effects of reduced redeemability on peer-to-peer token inflows and outflows. In the short term, both inflows and outflows increased after redeemability was reduced. In the longer term, however, both inflow and outflows persistently declined to a level that is lower than the initial level prior to the event. Furthermore, there is no detectable difference in the decline in inflow and outflow levels between users with different redemption exposure.

Appendix Figure C4 shows the heterogeneous impact on token issuance by Bunz HQ, revealing that it also decreased significantly following the reduction in redeemability. Once again, this decrease did not differ across users with varying redemption exposure.

Appendix Figure C5 documents the heterogeneous impact on token holdings, showing that holdings decreased significantly after the redemption reduction. Consistent with larger increases in redemption among users closer to redemption opportunities, token holdings dropped more sharply for users with higher redemption exposure.

Second, we study how reduced redeemability affected token acceptance and redemption for users with different exposure to food and retail merchants, respectively. Specifically, we estimate the following specification:

$$y_{i,t} = \sum_{k=-11}^{15} (\beta_{I,k} I_{t-t^*=k} + \beta_{f,k} Exposure_{f,i} \times I_{t-t^*=k} + \beta_{r,k} Exposure_{r,i} \times I_{t-t^*=k}) + \delta_i + \varepsilon_{i,t}. \quad (8)$$

Appendix Figure C6 reports estimates of the above equation with token acceptance as the outcome variable, revealing that the reduction in token adoption is not accounted for by users initially closer to food merchants nor by those initially closer to retail merchants.

Appendix Figure C7 shows results for token redemption as the outcome variable, revealing that the spike in redemption is driven by users initially closer to food merchants, while users initially closer to retail merchants did not behave differentially. This finding is consistent with the fact that Bunz HQ only allowed redemption for the food merchant after redeemability was reduced.

6 The Heterogeneous Effects of Redemption Halt

In this section, we provide a second test of Prediction 2—namely, that a halt in redeemability may cause a large decline in token acceptance due to a transition from a monetary equilibrium to a non-monetary one. To do so, we estimate the heterogeneous effects of the final halt in token redemption on users with different exposure to redemption opportunities. Since this event is contaminated by the subsequent arrival of Covid-19, the estimates here are more descriptive than causal. The estimates are once again suggestive that changes in redeemability can result in changes in token acceptance that are much larger than initial cross-sectional differences.

6.1 Methodology

The final suspension of the Shop Local program was announced on February 28, 2020. The halt was unexpected and, unlike the previous partial reduction, was attributed to temporary technical difficulties. Despite this official statement, the program was never restarted. As previously shown in Figure 2, token redemption immediately and permanently ceased following the suspension.

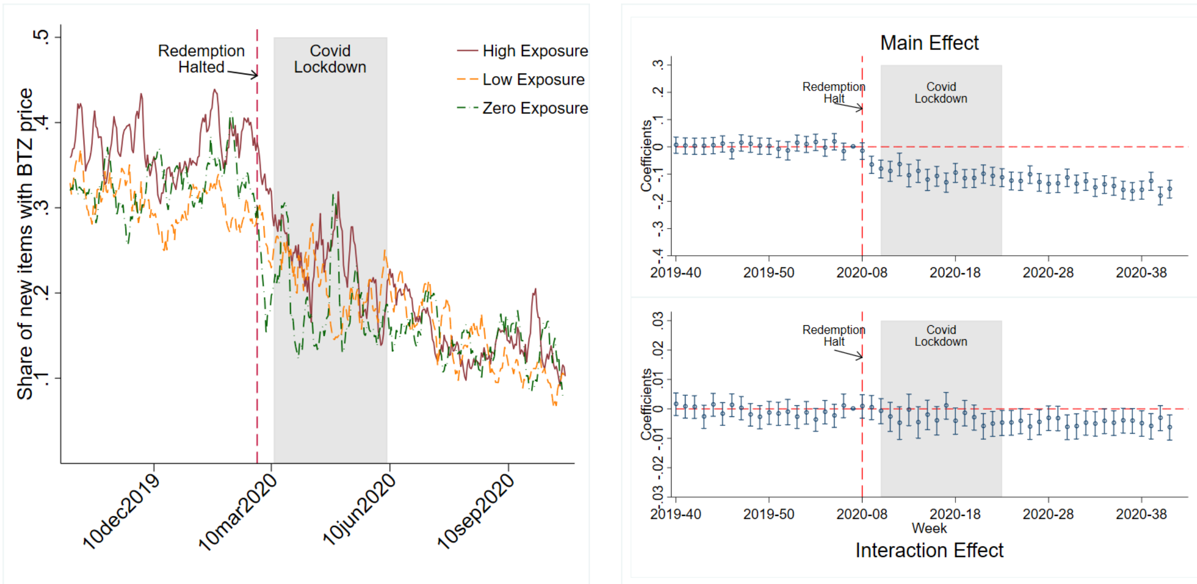
Assessing the long-term trade impact of the redemption halt is confounded by the onset of the Covid-19 lockdown in Toronto on March 12, 2020. However, the high-frequency nature of our data allows us to disentangle these two effects. Following previous sections, we first plot the trend of the token acceptance of frequent users with different exposure.[19] Then, we estimate the effects of the redemption halt on active users using an interrupted time series

[19] We further show the trend of token inflow, outflow, issuance, and holdings in Appendix C.

Figure 7. Effects of redemption halt on token acceptance

Panel A: Daily trend

Panel B: User-level estimates



Notes: The figure plots the effect of redemption halt on token acceptance. Panel A shows the 7-day moving average share of new items with a posted BTZ price by redemption exposure. Panel B plot the regression coefficients $\beta_{I,k}$ and $\beta_{D,k}$, with the 95 percent confidence intervals, estimated by Equation (9) using the share of new items with a posted BTZ price as outcome variable. The red dashed line indicates February 28, the day of full cessation of Shop Local program.

design. Specifically, we construct a weekly panel of active users, spanning the 20 weeks before the event until 34 weeks after, and estimate the following regression equation:

$$y_{i,t} = \sum_{k=-20}^{34} (\beta_{I,k} I_{t-t^*=k} + \beta_{D,k} Exposure_i \times I_{t-t^*=k}) + \delta_i + \varepsilon_{i,t}, \tag{9}$$

where $y_{i,t}$ denotes observed outcomes of user i in week t, including token redemption, acceptance, flows, and holdings, $I_{t-t^*=k}$ are dummies indicating each week relative to when redeemability was reduced, $Exposure_i$ is the the number of merchants within 1 km of users, δ_i are user fixed effects, and $\varepsilon_{i,t}$ is an error term. Standard errors are clustered at user level.

6.2 Results

Figure 7 confirms that token acceptance dramatically fell following redemption halt, while displaying the main result of this section—namely, that the impact on token acceptance varied with user initial redemption exposure only in a very limited manner.

Panel A shows that token acceptance—as measured by the share of item posts with a BTZ price—immediately decline from roughly 30 p.p to roughly 23 p.p. after redeemability was halted. Thereafter, the share continued a gradual downward trend, unaffected by the onset of Covid-19. By the end of 2020, only about 10 percent of item posts carried a BTZ price. The figure also shows that the initial heterogeneity among user groups gradually dissipated over time. Some initial difference persistent even at the beginning of Covid-19. After the Covid-19 lockdown, however, there was virtually no difference in token acceptance among users with different redemption exposure.

Panel B confirms that token acceptance dropped immediately by more than 10 pp. Meanwhile, there is little immediate differential impact of the redemption halt according to initial redemption exposure on token acceptance. A small and statistically significant difference emerged only 14 weeks after the redemption halt.

6.3 Discussion and Additional Results

The above estimates confirm that the redemption halt precipitated a dramatic and persistent decline in token acceptance, as previously shown by Wong (2025). Additionally, the results reveal that this collapse was largely uniform; the impact of the halt varied only minimally with users' initial redemption exposure. While some cross-sectional differences persisted initially, they gradually dissipated, resulting in virtually no difference in token acceptance across user groups after the COVID-19 lockdown. These findings suggest that the aggregate shock of redemption halt overwhelmed individual incentives related to redemption exposure, causing a system-wide collapse in token acceptance.

Several additional results enrich the above analysis.

Appendix Figures D1 and D2 illustrate the heterogeneous effects of the redemption halt on peer-to-peer token flows. The redemption halt precipitated a large drop in inflows and outflows that persisted through the COVID-19 lockdown, mirroring the sustained decline in token acceptance. Furthermore, the gap in token inflows between users with differing redemption exposures narrowed immediately after the halt and did not recover post-lockdown. Notably, the decline in token outflow after the redemption halt was indistinguishable across users with varying levels of exposure.

Appendix Figure D3 depicts the impact on token issuance by Bunz HQ, which fell significantly following the redemption halt and continued to decline after the COVID-19 lockdown. While the immediate reduction following the program halt did not vary by user exposure, the subsequent decline post-lockdown was heterogeneous: token issuance toward users with higher redemption exposure experienced a significantly larger drop.

Appendix Figure D4 documents the impact on token holdings, showing that holdings increased significantly after the redemption halt. Once again, this increase did not differ across users with varying redemption exposure.

Second, we study how redemption halt affected token acceptance for users with different exposure to food and retail merchants, respectively. Specifically, we estimate the following specification:

$$y_{i,t} = \sum_{k=-20}^{34} (\beta_{I,k} I_{t-t^*=k} + \beta_{f,k} Exposure_{f,i} \times I_{t-t^*=k} + \beta_{r,k} Exposure_{r,i} \times I_{t-t^*=k}) + \delta_i + \varepsilon_{i,t}. \quad (10)$$

Appendix Figure D5 reports estimates of the above equation with token acceptance as the outcome variable, revealing that the reduction in token adoption is not accounted for by users initially closer to food merchants nor by those initially closer to retail merchants.

7 Optimal Currency Redemption

Having shown the empirical relevance of the heterogeneous-agent model (via empirical tests of Predictions 1 and 2), we now briefly analyze optimal redemption policy from the perspective of the currency issuer in the heterogeneous-agent model. Our key finding is that it may be optimal for the currency issuer to restrict redemption.

We consider the following game. First, the currency issuer chooses redemption utility

ν_R^i for each agent i. We assume that issuance σ is chosen such that the money stock M is in steady state. Second, agents simultaneously choose π_i and ρ_i to maximize steady-state payoffs. We assume that the currency issuer knows the distribution of u_C^i, but does not observe each agent's individual u_C^i.

We focus on redemption policies where the issuer randomly assigns a share s of agents a positive redemption utility given by $\nu_R = \frac{c}{\beta}(1+\sigma\beta)+\epsilon$, where ϵ is a small positive number indicating that the utility is just above the cutoff value $\frac{c}{\beta}(1+\sigma\beta)$. The remaining share $1-s$ of agents receive zero redemption utility. As discussed in Lemma 1 and illustrated in Figure 4, agents are guaranteed to accept money—that is, to take action $\pi = 1$—if and only if they are assigned a redemption utility of at least $\frac{c}{\beta}(1+\sigma\beta)$. Conditional on accepting money, further increases in redemption utility lead to a higher likelihood that the agent will redeem the money, i.e., take action $\rho = 1$, which in turn raises the redemption cost. Randomization is without loss, since the issuer does not know the individual u_c^i of agents.

The issuer's objective function is given by

$$\max_s \underbrace{\tau l M W}_{\text{Revenue}} - \underbrace{\int \mu^i \rho^i \nu_R^i di}_{\text{Redemption cost}} \tag{11}$$

where τ is a transaction fee charged by the issuer per monetary transaction. We assume that the fee is imposed on the buyer. By substituting in the definitions of W, M, and Equation (19), the equilibrium conditions are then given by

$$W = \frac{lM+\sigma}{lM+\sigma+1} s Pr(u_c^i - \tau < e(W)) + \frac{lM+\sigma}{lM+\sigma+lW} Pr(u_c^i - \tau > e(W)) \tag{12}$$

$$M = \frac{\sigma}{\sigma+lW}(1-s) Pr(u_c^i - \tau < e(W)) + \frac{lM+\sigma}{lM+\sigma+lW} Pr(u_c^i - \tau > e(W)). \tag{13}$$

The solutions to these fixed-point equations determine the issuer's steady-state revenue, which is given by τlMW. The issuer's redemption cost is

$$\text{Redemption cost} = \int \mu^i \rho^i \nu_R^i di = s\nu_R \frac{lM + \sigma}{lM + \sigma + 1} Pr(u_c^i - \tau < e(W)). \tag{14}$$

Although we do not have an analytical proof of equilibrium uniqueness, our simulations using multiple initial guesses suggest that the solution is unique under plausible parameter values.

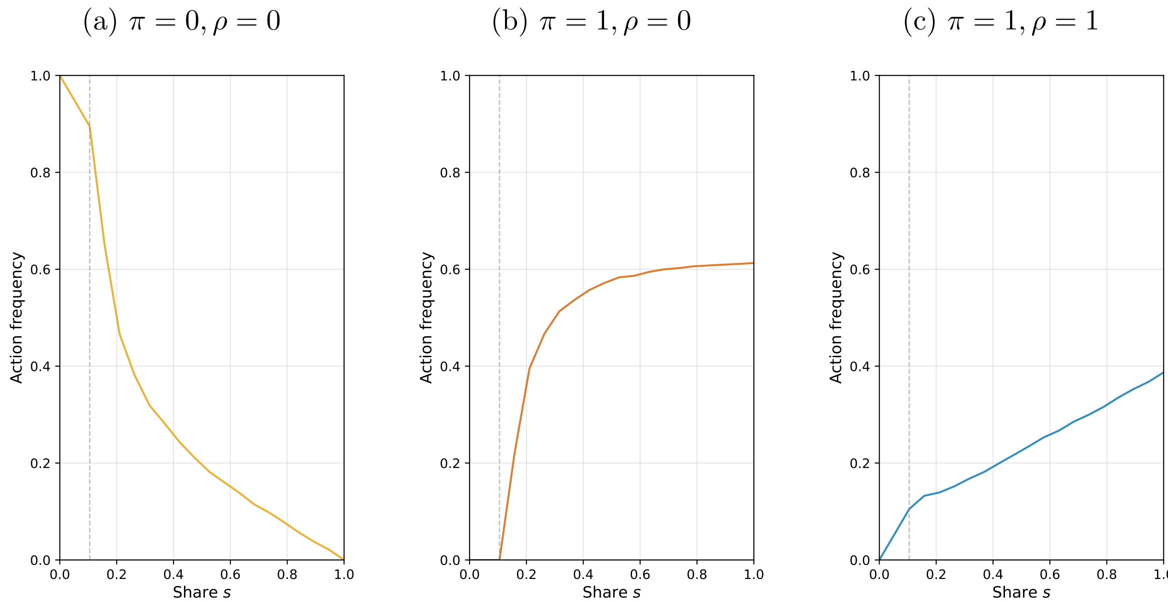
Our main finding is that issuer profits may be non-monotone in redemption share s. To see this, consider how agent action profiles change as the redemption share s increases. When $s = 0$, there is a unique non-monetary equilibrium, in which all agents choose $\pi = 0, \rho = 0$ (i.e., don't accept, don't redeem), as previously shown in Proposition 3. As the issuer increases s, agents begin to accept money for the purpose of redemption, leading them to choose $\pi = 1, \rho = 1$ (i.e., accept and redeem). As s continue to increase, the measure of agents willing to accept money may exceed a critical mass, such that the transaction value of money is sufficient to motivate agents with high u_C^i to accept money purely for transaction purposes, so some agents begin to choose $\pi = 1, \rho = 0$ (i.e., accept and don't redeem).

Figure 8 uses a numerical simulation to illustrate how the share of agents with different action profiles varies with redemption share s. Panel (a) shows that the share of agents who neither accept nor redeem decreases as the redemption share s increases. Panel (b) shows that the share of agents who accept but don't redeem is initially zero when s is small, but starts to rapidly rise once s exceeds a threshold. The cutoff in s corresponds to the point at which the measure of accepting agents W exceeds the critical mass needed to incentivize the agent with the highest u_C^i to accept money for transaction purposes. Panel (c) shows that the share of agents who accept and redeem starts at zero and increases steadily with s.[20]

Figure 9 shows how revenue, cost, and profit vary with s. Panel (a) shows that revenue initially increases with s but eventually declines. Panel (b) shows that cost rises with s, but the marginal cost falls around the critical threshold where the frequency of $\pi = 1, \rho = 0$

[20] Figure 8 omits the action frequency of $\pi = 1, \rho = 0$ because it remains constant at 0, given that the redemption utility we calibrate is $\nu_R^i \equiv \frac{c}{\beta} + \sigma + \epsilon$.

Figure 8. Action frequency vs. share of agents given redemption opportunity

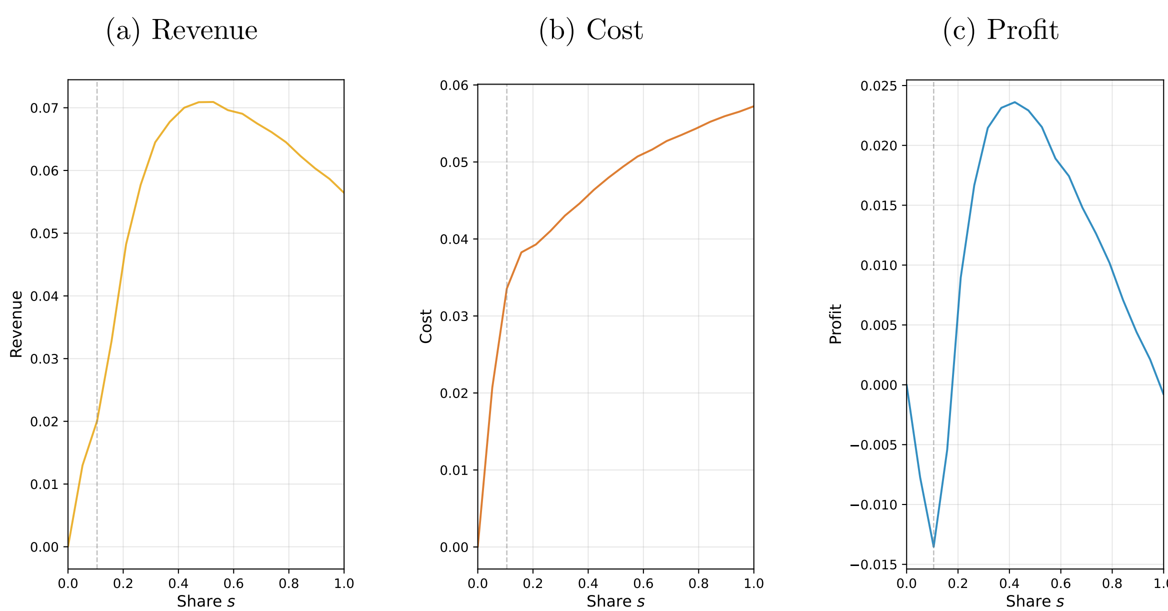


Notes. This figure shows the equilibrium action frequencies as a function of the redemption assignment share s. The dashed line indicates the cutoff s that induces positive action frequency for $\pi = 1, \rho = 0$. The action frequency of $\pi = 0, \rho = 1$ is a constant 0 and hence omitted from the figure. The parameters used in simulation are $\tau = 1$, $l = 0.5$, $\beta = 0.9$, $c = 1$, $\sigma = 0.05$, $u_C \in [1, 5]$, $\nu_R = 1.39 = \frac{c}{\beta}(1 + \beta\sigma) \times 120\%$.

becomes positive. Panel (c) shows that the profit-maximizing redemption share s is interior.

The intuition for Figure 9 comes directly from the action profiles. As s increases from zero, revenue-generating transactions are initially driven entirely by agents with action profiles $\pi = 1, \rho = 1$—those who accept money solely for the purpose of immediate redemption—resulting in high marginal costs for the issuer. Once s exceeds the critical threshold, the frequency of the action $\pi = 1, \rho = 0$ begins to rise, which increases marginal revenue and reduces marginal cost, pushing the issuer toward its profit-maximizing share. However, as s continues to increase, the frequency of $\pi = 1, \rho = 0$ plateaus, while the frequency of $\pi = 1, \rho = 1$ continues to rise, leading to higher costs and ultimately lower revenue and profits.

Figure 9. Revenue, cost, profit vs. share of agents given redemption opportunity



Notes. This figure shows the equilibrium revenue, cost, and profit as a function of the redemption assignment share s. The dashed line indicates the cutoff s that induces positive action frequency for $\pi = 1, \rho = 0$. The parameters used in simulation are $\tau = 1$, $l = 0.5$, $\beta = 0.9$, $c = 1$, $\sigma = 0.05$, $u_C \in [1, 5]$, $\nu_R = 1.39 = \frac{c}{\beta}(1 + \beta\sigma) \times 120\%$.

8 Conclusion

Redemption is central to the successful circulation of many currencies and payment tools. However, rigorous economic analysis of the impact of currency redemption policy has been rare, partly because of the lack of appropriate data. This paper leverages both cross-sectional and time-series variation in redemption convenience from the Bunz setting to estimate the impacts of redemption policy. Our key finding is that redemption promises serve as more than an implicit backing for the value of a currency; they affect the cross-sectional distribution of money acceptance as well as money and good flows. We find that redemption convenience increases users' desire to accept a new currency, leading token to flows on average towards those closer to redemption opportunities. We also find that the removal of redeemability triggered a collapse in token acceptance everywhere, providing clear evidence of strategic complementarity in token acceptance.

We interpret these results through the lens of a search-theoretic model with endogenous redemption choices. We show that a credible promise of redemption can coordinate agents on a unique monetary equilibrium at *zero steady-state cost*, even as it introduces the possibility of a run equilibrium. We also show in a heterogeneous-agent model with partial acceptability—which better explains the evidence—that optimal redemption policy involves some degree of redemption frictions. These findings can help inform the design and roll-out of new digital currencies. Since prior studies have largely focused on optimal adoption subsidies and issuance policies, a better understanding of the impacts of currency redemption policy helps to maximize efficiency and mitigate systemic risks.

References

Aiyagari, S.Rao, and Neil Wallace. 1997. "Government Transaction Policy, the Medium of Exchange, and Welfare." *Journal of Economic Theory* 74, 1–18.

Alvarez, Fernando, David Argente, and Diana Van Patten. 2023. "Are cryptocurrencies currencies? Bitcoin as legal tender in El Salvador." *Science* 382 eadd2844.

Alvarez, Fernando E, David Argente, Francesco Lippi, Esteban Méndez, and Diana Van Patten. 2023. "Strategic Complementarities in a Dynamic Model of Technology Adoption: P2P Digital Payments."Technical report, National Bureau of Economic Research.

Berentsen, Aleksander, Gabriele Camera, and Christopher Waller. 2007. "Money, credit and banking." *Journal of Economic Theory* 135, 171–195.

Chiu, Jonathan, Seyed Mohammadreza Davoodalhosseini, Janet Jiang, and Yu Zhu. 2023. "Bank Market Power and Central Bank Digital Currency: Theory and Quantitative Assessment." *Journal of Political Economy* 131, 1213–1248.

Cong, Lin William, Ye Li, and Neng Wang. 2022. "Token-based Platform Finance." *Journal of Financial Economics* 144, 972–991.

Crouzet, Nicolas, Apoorv Gupta, and Filippo Mezzanotti. 2023. "Shocks and Technology Adoption: Evidence from Electronic Payment Systems." *Journal of Political Economy* 131, 000–000.

Diamond, Douglas W., and Philip H. Dybvig. 1983. "Bank Runs, Deposit Insurance, and Liquidity." *Journal of Political Economy* 91, 401–419. 10.1086/261155.

Friedman, Milton, and Anna Jacobson Schwartz. 2008. *A Monetary History of the United States, 1867-1960*. Volume 9. Princeton University Press.

Garratt, Rodney J., and Maarten R. C. van Oordt. 2024. "Crypto Exchange Tokens."Technical report, Bank for International Settlements.

Gorton, Gary. 1996. "Reputation Formation in Early Bank Note Markets." *Journal of Political Economy* 104, 346–397.

Graeber, David. 2011. *Debt: The First 5000 Years*. Brooklyn, N.Y. : Melville House.

Green, Edward J., and Ruilin Zhou. 1998. "A Rudimentary Random-Matching Model with Divisible Money and Prices." *Journal of Economic Theory* 81, 252–271. https://doi.org/10.1006/jeth.1997.2356.

Green, Edward J., and Ruilin Zhou. 2002. "Dynamic Monetary Equilibrium in a Random Matching Economy." *Econometrica* 70, 929–969.

Gu, Chao, Cyril Monnet, Ed Nosal, and Randall Wright. 2023. "Diamond–Dybvig and beyond: On the instability of banking." *European Economic Review* 154 104414.

Hamilton, Earl J. 1946. "The First Twenty Years of the Bank of Spain. I." *Journal of Political Economy* 54, 17–37.

Huang, Xuesong, and Todd Keister. 2025. “Can Redemption Fees Prevent Runs on Funds?” *SSRN*. 10.2139/ssrn.4873088, Last revised: July 1, 2025.

Humphrey, Caroline. 1985. “Barter and Economic Disintegration.” *Man* 20, 48–72.

Jean, Kasie, Stanislav Rabinovich, and Randall Wright. 2010. “On the multiplicity of monetary equilibria: Green–Zhou meets Lagos–Wright.” *Journal of Economic Theory* 145, 392–401.

Kamiya, Kazuya, and Takashi Shimizu. 2006. “Real indeterminacy of stationary equilibria in matching models with divisible money.” *Journal of Mathematical Economics* 42, 594–617.

Kamiya, Kazuya, and Takashi Shimizu. 2007a. “On the Role of Tax Subsidy Scheme in Money Search Models.” *International Economic Review* 48, 575–606.

Kamiya, Kazuya, and Takashi Shimizu. 2007b. “Existence of Equilibria in Matching Models of Money: A New Technique.” *Economic Theory* 32, 447–460. 10.1007/s00199-006-0135-1.

Kamiya, Kazuya, and Takashi Shimizu. 2011. “Stationary monetary equilibria with strictly increasing value functions and non-discrete money holdings distributions: An indeterminacy result.” *Journal of Economic Theory* 146, 2140–2150.

Kiyotaki, Nobuhiro, and Randall Wright. 1993. “A Search-theoretic Approach to Monetary Economics.” *The American Economic Review*, 63–77.

Knapp, Georg Friedrich. 1924. *The State Theory of Money*. McMaster University Archive for the History of Economic Thought.

Lerner, Abba P. 1947. “Money as a Creature of the State.” *The American Economic Review* 37, 312–317.

Li, Yiting, and Randall Wright. 1998. “Government Transaction Policy, Media of Exchange, and Prices.” *Journal of Economic Theory* 81, 290–313.

Liu, Jiageng, Igor Makarov, and Antoinette Schoar. 2023. “Anatomy of a Run: The Terra Luna Crash.”Technical report, National Bureau of Economic Research.

Lotz, Sebastien. 2004. “Introducing a new currency: Government policy and prices.” *European Economic Review* 48, 959–982.

Lotz, Sebastien, and Guillaume Rocheteau. 2002. “On the Launching of a New Currency.” *Journal of Money, Credit and Banking* 34, 563–588.

Ma, Yiming, Yao Zeng, and Anthony Lee Zhang. 2025. “Stablecoin Runs and the Centralization of Arbitrage.” *Review of Financial Studies*, `https://sfs.org/rfs-forthcoming-paper-85/`, Forthcoming.

Mitchell-Innes, A. 1913. *What is Money?*. Cosimo Classics.

Mitchell-Innes, A. 1914. *The Credit Theory of Money*. Cosimo Classics.

Rocheteau, Guillaume, Pierre-Olivier Weill, and Tsz-Nga Wong. 2021. "An heterogeneous-agent New-Monetarist model with an application to unemployment." *Journal of Monetary Economics* 117, 64–90.

Rogoff, Kenneth S, Zhiheng He, and Yang You. 2024. "Market Power and Redeemable Loyalty Token Design."Technical report, National Bureau of Economic Research.

Rogoff, Kenneth, and Yang You. 2023. "Redeemable Platform Currencies." *The Review of Economic Studies* 90, 975–1008.

Sanches, Daniel. 2016. "The Free-Banking Era: A Lesson for Today." *Economic Insights. Federal Reserve Bank of Philadelphia Research Department*, 9–14.

Selgin, George. 2003. "Adaptive Learning and the Transition to Fiat Money." *The Economic Journal* 113, 147–165.

Shevchenko, Andrei, and Randall Wright. 2004. "A Simple Search Model of Money with Heterogeneous Agents and Partial Acceptability." *Economic Theory* 24, 877–885.

Sockin, Michael, and Wei Xiong. 2023. "A model of cryptocurrencies." *Management Science* 69, 6684–6707.

Soller Curtis, Elisabeth, and Christopher J. Waller. 2000. "A search-theoretic model of legal and illegal currency." *Journal of Monetary Economics* 45, 155–184.

Velde, François R. 2007. "John Law's System." *American Economic Review* 97, 276–279.

Volta, Richard Dalla. 1893. "The Italian Banking Crisis." *Journal of Political Economy* 2, 1–25.

Williamson, Stephen. 2024. "Deposit insurance, bank regulation, and narrow banking." *Journal of Economic Theory* 219 105859.

Wong, Michael B. 2025. "Money and Barter in the Field: Evidence from a Digital Currency Experiment." Working paper.

Wray, L. Randall. ed. 2004. *Credit and State Theories of Money.* Edward Elgar Publishing.

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A Proofs

Proof of Proposition 1. The proof strategy follows the framework of Kiyotaki and Wright (1993). We first observe that n equilibrium, the measure of agents choosing not to redeem upon receiving one unit of money must be equal to the money supply. Hence, μ_1 and M satisfy $\mu_1(1-\rho) = M$. Steady state requires that the flows of agents between state 0 and state 1 are equalized, such that

$$\overbrace{\mu_1}^{\text{Prob. of being in S1}} \cdot \underbrace{[\rho + (1-\rho)l\mathit{\Pi}(1-\mu_1)]}_{\text{Prob. of transition, S1 to S0}} = \overbrace{(1-\mu_1)}^{\text{Prob. of being in S0}} \cdot \underbrace{lM\pi.}_{\text{Prob. of transition, S0 to S1}} \tag{15}$$

Next, note that the value functions are linear in decision variables ρ, π. Let $\mathit{\Delta} = V^1 - V^0$, we have

$$\mathit{\Delta} = V^1 - V^0 = \frac{\rho\nu_R + (1-\rho)l\mathit{\Pi}(1-\mu_1)u_C + \pi lMc}{1 - \beta(1-\rho)(1 - l\mathit{\Pi}(1-\mu_1)) + \beta\pi lM}. \tag{16}$$

By concentrating out the linear coefficients of decision variables, we find that individual optimal solutions are pure strategies with $\pi, \rho \in \{0, 1\}$, given by

$$\rho = 1 \;\Leftrightarrow\; \mathit{\Delta} \leq \frac{\nu_R - l\mathit{\Pi}(1-\mu_1)u_C}{\beta(1 - l\mathit{\Pi}(1-\mu_1))} \tag{17}$$

$$\pi = 1 \;\Leftrightarrow\; \mathit{\Delta} \geq \frac{c}{\beta} \tag{18}$$

which constitute a fixed point system.

There are four potential pure-strategy symmetric steady-state equilibria.

1. All agents play $\rho = 0, \pi = 0$. In this case, $\mathit{\Delta} = 0$. The individual optimality constraints are $\mathit{\Delta} \leq \frac{c}{\beta}$ and $\mathit{\Delta} \geq \frac{\nu_R}{\beta}$. The solution region is given by $\nu_R \leq 0$.

2. All agents play $\rho = 1, \pi = 0$. In this case, $\mathit{\Delta} = \nu_R$ and $\mu_1 = M = 0$. The individual optimality constraints are $\mathit{\Delta} < \frac{c}{\beta}$ and $\mathit{\Delta} \leq \frac{\nu_R}{\beta}$. The solution region is given by $0 \leq \nu_R \leq \frac{c}{\beta}$.

3. All agents play $\rho = 1, \pi = 1$. In this case, $\mu_1 = M = 0$ and $\mathit{\Delta} = \nu_R$. The individual optimality constraints are $\mathit{\Delta} \geq \frac{c}{\beta}$ and $\mathit{\Delta} \leq \frac{\nu_R - lu_C}{\beta(1-l)}$. The solution region is given by $\nu_R \geq \max\{\frac{c}{\beta}, \overline{\nu}\}$, where $\overline{\nu} \equiv \frac{lu_c}{1-\beta(1-l)}$.

4. All agents play $\rho = 0, \pi = 1$. In this case, $\mathit{\Delta} = \frac{l(1-M)u_C + lMc}{1-\beta(1-l(1-M))+\beta lM}$. The individual optimality constraints are $\mathit{\Delta} \geq \frac{c}{\beta}$ and $\mathit{\Delta} > \frac{\nu_R - l(1-M)u_C}{\beta(1-l(1-M))}$. The solution region is given by $\nu_R < \bar{\nu}(M) = \frac{l(1-M)u_C + \beta lMc + \beta l^2 M(1-M)(u_C - c)}{1-\beta(1-l(1-M))+\beta lM}$, conditional on a parametric restriction that $u_C > \underline{u} = \frac{c(1-\beta+\beta l(1-M))}{\beta l(1-M)}$.

Note that

$$\frac{\partial\bar{\nu}}{\partial x} = \text{PositiveConst} \cdot \big(\underbrace{-u_C + \beta c + \beta l(u_C - c)(2x-1)}_{:=g(x)}\big)$$

Since $u_C > c$, $g(x)$ is increasing in x, and $g(1) = \beta l(u_C - c) - (u_C - \beta c) < 0$ due to $u_C > c$. So $g(x) < 0$ for all $x \in [0, 1]$ so we have $\bar{\nu}(M) < \bar{\nu}(0)$ for all $M > 0$.

Proof of Lemma 1 (Optimal individual decisions). We begin by fully specifying the assumptions embedded in the model's transition dynamics. Note that individual state transition follows the law of motion,

$$\mu^i_{t+1} = \mu^i_t(1 - \rho^i_t)(1 - lW_t) + (1 - \mu^i_t)(\pi^i_t lM_t + \sigma). \tag{19}$$

We let agents' individual expectations satisfy $E[V^i_{k,t+1}] = V^i_{k,t}$ for $k \in \{0, 1\}, i \in \mathcal{I}$. In equilibrium, this assumption doesn't impose any additional restrictions. The economy initializes with individual and aggregate states $\{\mu^i_1\}_i, W_1, M_1$. At the beginning of period t, agents observe the strategy profiles in period $t - 1$, which is equivalent to observing $\{\mu^i_t\}_i, M_t, W_t$. The dynamic path towards equilibrium is iterated forward as the following: initialize with $\{\mu^i_1\}_i, W_1, M_1$; agents best respond and generate $\{\pi^i_1\}_i, \{\rho^i_1\}_i$; individual and aggregate states are updated overnight and results in $\{\mu^i_2\}_i, W_2, M_2$; agents best respond and generate $\{\pi^i_2\}_i, \{\rho^i_2\}_i$; and this process continues iteratively for all subsequent periods.

Based on the Bellman equations, first order conditions of individual choices π^i_t, ρ^i_t are given by

$$FOC_{\pi^i_t} = lM_t\left(\beta(V^i_{1,t} - V^i_{0,t}) - c\right)$$
$$FOC_{\rho^i_t} = \nu^i_R - lW_t u^i_C - \beta(1 - lW_t)(V^i_{1,t} - V^i_{0,t})$$

Optimal decisions are corner solutions in $\{0, 1\}$ depending on the sign of the first order condition. We specify the following tie-breaking rules. If $FOC_{\pi^i_t} = 0$, then the optimal $\pi^i_t = 1$. If $FOC_{\rho^i_t} = 0$, then the optimal $\rho^i_t = 0$.

Hence we can characterize the optimal decision rule as the following. Let

$$\varDelta^i_t = V^i_{1,t} - V^i_{0,t} = \frac{\rho^i_t \nu^i_R + (1 - \rho^i_t) lW_t u^i_C + \pi^i_t lM_t c}{1 - \beta(1 - \rho^i_t)(1 - lW_t) + \beta\pi^i_t lM_t + \beta\sigma} \tag{20}$$

and $\pi^i_t = 1$ iff $\varDelta^i_t \geq \frac{c}{\beta}$, $\rho^i_t = 1$ iff $\varDelta^i_t < \frac{\nu^i_R - lW_t u^i_C}{\beta(1-lW_t)}$. We stack these conditions in Appendix Table 1 "Conditions" column.

To characterize optimal decisions π^i_t, ρ^i_t as functions of ν^i_R, u^i_C, we first derive simplifications of the optimality conditions. We do so by defining

$$\boxed{u} := \frac{lW_t u_C}{1 - \beta(1 - lW_t) + \beta\sigma} \tag{37}$$

and

$$u := \frac{\beta(1 - lW_t)lM_t c + (1 + \beta lM_t + \beta\sigma)lW_t u_C}{1 - \beta(1 - lM_t - lW_t)} \tag{38}$$

and deriving column "Simplified" in Appendix Table 1 that is equivalent to column "Conditions."

Equil.	Δ_t^i	Conditions	Simplified
$\pi_t^i = 0,$ $\rho_t^i = 0$	$\Delta_t^i = \frac{lW_t u_C}{1-\beta(1-lW_t)+\beta\sigma}$	$\frac{\nu_R^i - lW_t u_C}{\beta(1-lW_t)} \leq \frac{lW_t u_C}{1-\beta(1-lW_t)+\beta\sigma}$ (21) $\frac{lW_t u_C}{1-\beta(1-lW_t)+\beta\sigma} < \frac{c}{\beta}$ (22)	$\frac{\nu_R^i}{1+\sigma\beta} \leq \boxed{u}$ (23) $\boxed{u} < \frac{c}{\beta}$ (24)
$\pi_t^i = 1,$ $\rho_t^i = 0$	$\Delta_t^i = \frac{lW_t u_C + lM_t c}{1-\beta(1-lW_t)+\beta lM_t+\beta\sigma}$	$\frac{\nu_R^i - lW_t u_C}{\beta(1-lW_t)} \leq \frac{lW_t u_C + lM_t c}{1-\beta(1-lW_t)+\beta lM_t+\beta\sigma}$ (25) $\frac{c}{\beta} \leq \frac{lW_t u_C + lM_t c}{1-\beta(1-lW_t)+\beta lM_t+\beta\sigma}$ (26)	$\nu_R^i \leq u$ (27) $\frac{c}{\beta} \leq \boxed{u}$ (28)
$\pi_t^i = 0,$ $\rho_t^i = 1$	$\Delta_t^i = \frac{\nu_R^i}{1+\beta\sigma}$	$\frac{\nu_R^i}{1+\beta\sigma} < \frac{\nu_R^i - lW_t u_C}{\beta(1-lW_t)}$ (29) $\frac{\nu_R^i}{1+\beta\sigma} < \frac{c}{\beta}$ (30)	$\boxed{u} < \frac{\nu_R^i}{1+\beta\sigma}$ (31) $\frac{\nu_R^i}{1+\beta\sigma} < \frac{c}{\beta}$ (32)
$\pi_t^i = 1,$ $\rho_t^i = 1$	$\Delta_t^i = \frac{\nu_R^i + lM_t c}{1+\beta lM_t+\beta\sigma}$	$\frac{\nu_R^i + lM_t c}{1+\beta lM_t+\beta\sigma} < \frac{\nu_R^i - lW_t u_C}{\beta(1-lW_t)}$ (33) $\frac{c}{\beta} \leq \frac{\nu_R^i + lM_t c}{1+\beta lM_t+\beta\sigma}$ (34)	$u < \nu_R^i$ (35) $\frac{c}{\beta} \leq \frac{\nu_R^i}{1+\beta\sigma}$ (36)

Table 1. Characterization of Individually Optimal Solutions

Next, we observe the following key property:

$$\boxed{u} < \frac{c}{\beta} \quad \Rightarrow \quad \frac{u}{1+\beta\sigma} < \frac{c}{\beta} \tag{39}$$

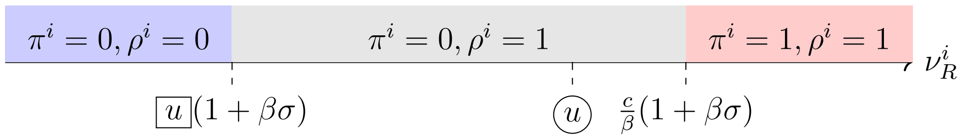
$$\boxed{u} \geq \frac{c}{\beta} \quad \Rightarrow \quad \frac{u}{1+\beta\sigma} \geq \frac{c}{\beta}. \tag{40}$$

This observation characterizes the relative tightness of inequalities in Appendix Table 1 "Simplified" column, and allows us to directly write down the structure of optimal decisions.

We are now ready to explicitly characterize optimal π_t^i, ρ_t^i as functions of ν_R^i, u_C^i. We separately characterize two possible cases of the agent.

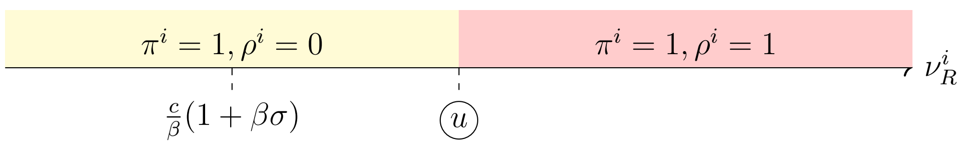
- First, suppose $\boxed{u} < \frac{c}{\beta}$, equivalently $u_C < e(W) = \frac{c}{\beta}\frac{1+\beta\sigma-\beta(1-lW)}{lW}$ indicating that transaction value of money alone is insufficient for agents to accept money.

  By Appendix Table 1 column "Simplified", the action pair $(\pi^i = 1, \rho^i = 0)$ cannot be optimal, and the remaining actions may be optimal. By Equation (39), we must have $u < \frac{c}{\beta}(1+\beta\sigma)$. Thus, the cross-sectional distribution of actions are characterized by the following cutoffs.

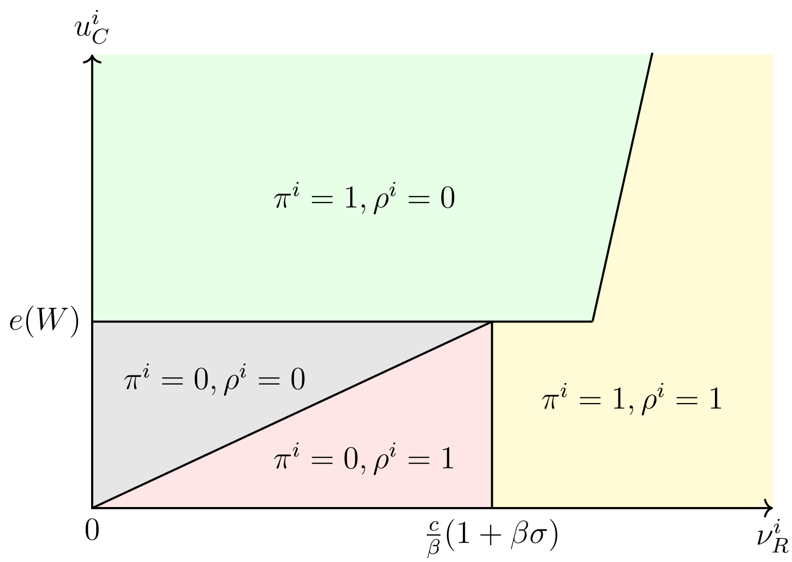


- Second, suppose $\boxed{u} \geq \frac{c}{\beta}$, equivalently $u_C \geq e(W)$, indicating that transaction value of money alone is high enough for agents to accept money.

  By Appendix Table 1, the action pair $(\pi^i = 0, \rho^i = 0)$ cannot be optimal. In addition, by Equation (40), the region that supports $(\pi^i = 0, \rho^i = 1)$ as optimal action is empty. Hence the remaining actions that may be optimal are $(\pi^i = 1, \rho^i = 0)$ and $(\pi^i = 1, \rho^i = 1)$. By Equation (40), we must have $u \geq \frac{c}{\beta}(1+\beta\sigma)$. Thus, the cross-sectional distribution of actions are characterized by the following cutoffs.



Combining the two cases produces exactly the result presented in Lemma 1: Given $W, M \in (0,1)$, each agent i's optimal choice (π^i, ρ^i) as a function of (u_C^i, ν_R^i) is given by the following figure.



Proof of Proposition 2. (1) is a direct result of Lemma 1. (2) and (3) follow from

$$S^i = lM\pi^i \frac{\rho^i + (1-\rho^i)}{\pi^i lM + \sigma + \rho^i + (1-\rho^i)} \tag{41}$$

$$P^i = lW(1-\rho^i)\frac{\pi^i lM + \sigma}{\pi^i lM + \sigma + \rho^i + (1-\rho^i)}. \tag{42}$$

Note that S^i, P^i are pinned down by the optimal decisions π^i, ρ^i. Hence, for agents i, j with $u_C^i = u_C^j, \nu_R^i < \nu_R^j$, it is straightforward to enumerate the different possibilities of their optimal decisions based on Lemma 1 and find that $S^i \leq S^j$ and $P^i \geq P^j$.

Proof of Proposition 3 This proof builds upon the proof of Lemma 1. Recall that the aggregate states are determined by

$$W_{t+1} = \int \pi_t^j (1 - \mu_t^j) dj = W_t \tag{43}$$

$$M_{t+1} = \int \mu_t^j (1 - \rho_t^j) dj = M_t \tag{44}$$

and

$$\mu_{t+1}^i = \mu_t^i (1 - \rho_t^i)(1 - lW_t) + (1 - \mu_t^i)(\pi_t^i l M_t + \sigma) = \mu_t \tag{45}$$

which can be rearranged as $\mu_t^i = \frac{\pi_t^i l M_t + \sigma}{\pi_t^i l M_t + \sigma + \rho_t^i + (1-\rho_t^i) l W_t}$.

Recall

$$e(W) = \frac{c}{\beta} \frac{1 + \beta\sigma - \beta(1 - lW)}{lW} \tag{46}$$

we know

$$e'(W) = -\frac{c}{\beta} \frac{1 - \beta(1 - \sigma)}{lW^2} < 0, \quad e''(W) = \frac{2c(1 - \beta(1 - \sigma))}{\beta l W^3} > 0 \tag{47}$$

and $e(0) = +\infty$, $e(1) = \frac{c(1-\beta(1-\sigma)+\beta l)}{\beta l}$.

Recall $u_C^i \sim U([\underline{u}_C, \overline{u}_C])$. Additionally we denote $\overline{W}$ s.t. $\overline{u}_C = e(\overline{W})$ and $\underline{W}$ s.t. $\underline{u}_C = e(\underline{W})$. Since e is monotonically decreasing in W, we know that $\overline{W} < \underline{W}$.

Assumption 1. $\underline{u}_C < e(1)$ *and* $\overline{u}_C > e(\frac{1}{1+\sigma})$.

Assumption 1 states that the dispersion of u_C^i among agents is large. The first cutoff is chosen to ensure that there exist some agents with low enough u_C^i such that they are unwilling to accept money even when all other agents are accepting money, and the second cutoff is chosen to ensure that there exist some agents with high enough u_C^i such that they accept money even if only a small share, $\frac{1}{1+\sigma}$, of other agents are accepting money. Intuitively, the existence of low u_C^i agents ensures that there *cannot* exist a monetary equilibrium without sufficient redemption. The existence of high u_C^i agents ensures that there *cannot* exist a non-monetary equilibrium when redemption is sufficiently high.

The first part assumes that $\underline{W} > 1$, equivalently, $e(1) > \underline{u}_C$ or $\underline{u}_C < \frac{c\left(1+\beta\sigma-\beta(1-l)\right)}{l\beta}$. Note that this is an assumption on primitives. The intuition is that we want there to be at least some people with low u_C that doesn't want to accept money for transaction's sake even if $W = 1$. This suggests that unless they have redemption utility, they won't join. This is important for eliminating the monetary equilibrium in the low-redemption regime. The second part assumes that $\overline{W} < \frac{1}{1+\sigma}$, equivalently $e(\frac{1}{1+\sigma}) < \overline{u}_C$. The intuition is that we want at least some people with high u_C such that they are happy to accept money for transaction's sake even if only $\frac{1}{1+\sigma}$ share of the population can accept their money.

Claim 1. If redemption is distributed as $\nu_R^i < \frac{c}{\beta} + \sigma$, the only equilibrium is $\pi^* = 0, \rho^* = 1$ for all agents, and aggregate $W^*, M^* = 0$.

Proof. Since $\nu_R^i < \frac{c}{\beta} + \sigma$, we can disaggregate our equilibrium condition to be

$$W = \frac{lW}{lM + \sigma + lW} P\Big(u_C^i > e(W)\Big) \tag{48}$$

$$M = \frac{lM + \sigma}{lM + \sigma + lW} P\Big(u_C^i > e(W)\Big) + \frac{\sigma}{\sigma + lW} P\Big(u_C^i \leq e(W), \nu_R^i < \boxed{u}^i(1 + \beta\sigma)\Big) \tag{49}$$

Observe immediately that $W^* = 0, M^* = 0$ satisfies the conditions and we have $e(0) = \infty$ meaning that $\pi^* = 0$ for everyone, which is consistent with the aggregates. We next show that there cannot be $W^* \in (0, 1)$ that solves the system.

Suppose $W \neq 0$, then we can simplify the W condition to be

$$\frac{1}{l} = \frac{1}{lM + \sigma + lW} P\Big(u_C^i > e(W)\Big) \tag{50}$$

by plugging this into the M condition, we obtain

$$0 = \frac{\sigma}{l} + \frac{\sigma}{\sigma + lW} P\Big(u_C^i \leq e(W), \nu_R^i < \boxed{u}^i(1 + \beta\sigma)\Big) > \frac{\sigma}{l} \tag{51}$$

which is a contradiction. This concludes the proof of Proposition 3 Part (1).

We next show a more general statement than Proposition 3 Part (2).

Claim 2. Under the following redemption regimes $\{\nu_R^i\}$, there exists a unique monetary equilibrium where all agents play $\pi^* = 1, \rho^* = 0$ or $\pi^* = 1, \rho^* = 1$, aggregate quantities satisfy $W^* \in (\overline{W}, \frac{1}{1+\sigma})$ and $M^* = 1 - (1 + \sigma)W^*$. The redemption regime must satisfy:

(1) $\nu_R^i \geq \frac{c}{\beta}(1 + \sigma\beta)$ for all i. This condition ensures that there is enough redemption incentive to encourage acceptance.

(2) $\nu_R^i \leq \max\left\{\frac{(1+\beta\sigma)l\overline{W}u_C^i}{1-\beta+l}, \frac{c}{\beta}(1 + \sigma\beta)\right\}$ for each i. This condition guarantees that a positive-measure set of agents play $(\pi^i = 1, \rho^i = 0)$ in equilibrium. It suggests that the platform should not allow people to redeem too much, especially should restrict redemption utility of those agents with high u_C^i, since they would have accepted money anyway.

It is straightforward to see that Proposition 3 Part (2) is a direct corollary of this claim, where we choose $\overline{\nu} = \frac{(1+\beta\sigma)l\overline{W}\overline{u}_C}{1-\beta+l}$.

Proof. Immediately note that any $W \leq \overline{W}$ cannot be an equilibrium given the redemption regimes. If $W \leq \overline{W}$, the redemption regimes guarantee that we shall have $u_C^i \leq e(W), \nu_R^i \geq \frac{c}{\beta}(1 + \sigma\beta)$ for all agents. Then, they optimally chooses $\pi = 1, \rho = 1$ and this results in aggregate $W' = \int \pi(1 - \mu) = \frac{1}{1+\sigma} > \overline{W}$ which is a contradiction.

Next, we want to show that there is a unique $W^* \in (\overline{W}, \frac{1}{1+\sigma})$ that solves the system.

We begin with the following observation. The maximal redemption bound given in the claim satisfies the following property, that is, for any i and for any potential equilibrium aggregates $W \in (\overline{W}, 1), M \in [0,1]$, we have

$$\frac{(1+\beta\sigma)l\overline{W}u_C^i}{1-\beta+l} \leq u^i = \frac{\beta(1-lW_t)lM_tc + (1+\beta lM_t+\beta\sigma)lW_tu_C^i}{1-\beta(1-lM_t-lW_t)} \tag{52}$$

which means that under the redemption regime, all agents have $\frac{c}{\beta}(1+\sigma\beta) \leq \nu_R^i \leq \max\{u^i, \frac{c}{\beta}(1+\sigma\beta)\}$ for all agents.

In this case, there can only be two optimal actions in equilibrium, either $\pi = 1, \rho = 0$ for agents with $u_C^i > e(W)$ because we know $\nu_R^i < u^i$; or $\pi = 1, \rho = 1$ for agents with $u_C^i < e(W)$ because we know $\nu_R^i > \frac{c}{\beta}(1+\sigma\beta)$.

Now our equilibrium conditions are given by

$$W = \frac{lW}{lM+\sigma+lW}P\Big(u_C^i > e(W)\Big) + \frac{1}{lM+\sigma+1}\Big(1-P\Big(u_C^i > e(W)\Big)\Big) \tag{53}$$

$$M = \frac{lM+\sigma}{lM+\sigma+lW}P\Big(u_C^i > e(W)\Big) \tag{54}$$

They can equivalently can be written as

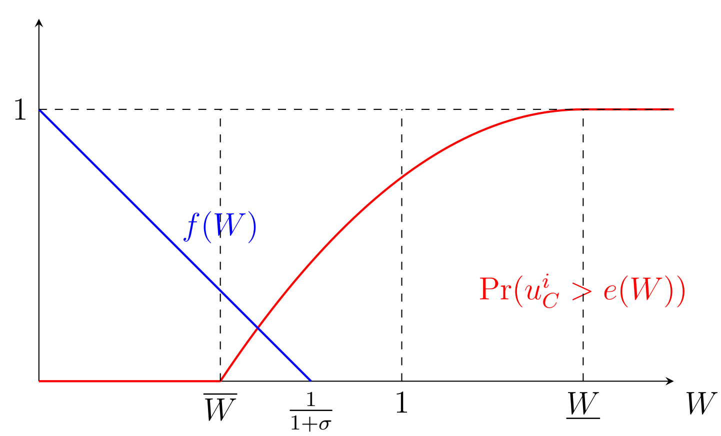
$$\frac{lM}{lM+\sigma}W + M = P\Big(u_C^i > e(W)\Big) \tag{55}$$

$$W = \frac{1-M}{1+\sigma} \tag{56}$$

It's easy to see that the first condition comes from the equation in M. The second condition can be obtained from taking the two equations in M, W and concentrating out the probability term.

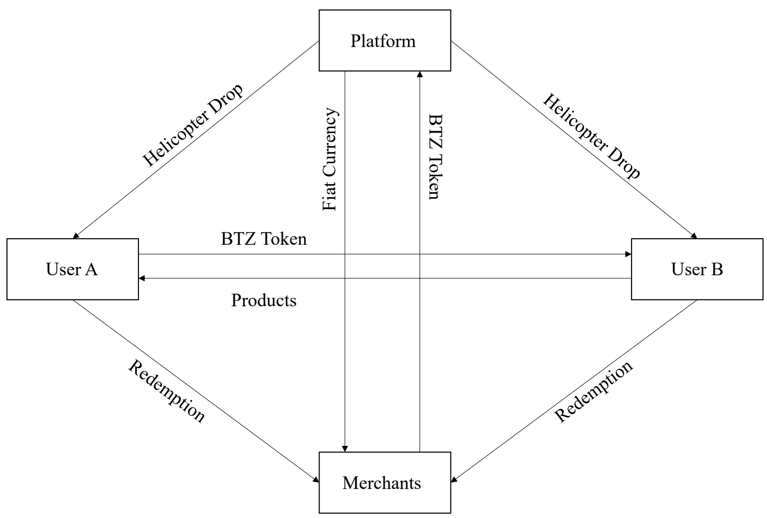
It is obvious that the second equation specifies a one-to-one mapping between equilibrium W^*, M^*, we plug this into the first eqation and simplifies the system to a single equation

$$f(W) \equiv -\sigma\frac{(W-\frac{1}{\sigma+1})(W-\frac{\sigma+l}{\sigma l})}{(W-\frac{\sigma+l}{(\sigma+1)l})} = P\Big(u_C^i > e(W)\Big). \tag{57}$$



A few additional observations regarding the left hand side function $f(W)$ yields our result. First note that the $\frac{1}{\sigma+1} < 1 < \frac{\sigma+l}{\sigma l} < \frac{\sigma+l}{(\sigma+1)l}$ and that $f(0) = 1$, $f(\frac{1}{\sigma+1}) = 0$. We also note that on the support $W \in [0,1]$, $f(W)$ is non-negative only on $[0, \frac{1}{1+\sigma}]$ and that on this region $f'(W) < 0$. Therefore we are guaranteed a unique solution to the equation such that $W \in (\overline{W}, \frac{1}{1+\sigma})$.

Figure A1. Token circulation within Bunz community



Notes: The figure plots the BTZ circulation in the BUNZ community.

Table A2. Examples of redeemable currency

Redeemable Currency Examples	Description	Redemption Policy and History
Bank Notes	The bank note is promoted by the tax authority enforcement. Specifically, a decree of October 7, 1716, ordered that the various tax collectors redeem the bank notes into cash on demand. On April 10, 1717, a decree made the bank notes explicit legal tender in the payment of taxes by individuals. The bank notes are backed by land, gold, and silver. The custodian and issuer of it is Banque Royale. The bank note circulated from 1716 to 1720.	The bank note is almost impossible to redeem. It collapsed in 1720 because of corruption, hyper-inflation, and misappropriation of money printing power.
US Dollar under Gold Standard	The Gold Reserve Act of 1934 made it illegal for private U.S. citizens to hold gold bullion (bars or coins) for monetary purposes. Individuals were required to surrender gold to the Federal Reserve in exchange for USD at a fixed rate of $35 per troy ounce. The USD is backed by gold. And the custodian and issuer is the Federal Reserve.	It is impossible for individuals and domestic institutions. A private citizen would have had no direct mechanism to redeem gold bars for dollars through official channels in the U.S. A foreign government or central bank in 1950 could redeem U.S. dollars for gold through the Federal Reserve at the official rate of $35 per ounce, as the U.S. was part of the Bretton Woods system (established in 1944). However, in 1971, President Richard Nixon ended the dollar's convertibility to gold, marking the definitive end of the global gold standard. The collapse is due to huge demand for the US dollar, money over-supply, and insufficient gold reserve.
Hong Kong Dollar	The HKD Pegged to GBP from 1935 to 1972, and to USD since 1983. The Hong Kong Monetary Authority is both the issuance entity and custodian of HKD. The HKD is backed by Great Britain Pounds and US dollar. The HKD help to achieve monetary sovereignty and flexibility despite Hong Kong's legal status as a Special Administrative Region of China.	The HKD is redeemable for US dollars through the banking system or currency OTC, but local payment with USD is not supported.
Q-coin (Gaming)	The Q-coin is a company guaranteed digital currency issued by Tencent backed by RMB.	The Q-coin can be redeemed in internet bars as OTC brokers, different brokers may charge other fees to intermediate Q-coin conversion to RMB. However, PBOC and the other six government administrations stopped Q-coin from being converted back to RMB and reiterated that Q-coins can only be used for gaming consumption. Still, payment for any physical goods is strictly prohibited. And the redemption system collpased in 2007.

Notes: This table reports the major redemption based currency issuance in history.

Examples of redeemable currency — Continued

Redeemable Currency Examples	Description	Redemption Policy and History
Wechat Pay Balance	The Wechat Pay Balance is introduced in 2013 by Tencent and backed by bank deposit. The redemption for Wechat Pay Balance is guaranteed by the Tencent and Bank Guarantee. A Regulator's Payment License is also needed to issue the digital money.	There is no cash withdrawal provided. Users can only redeem to their bank accounts. Additionally, there is no restriction set by the issuer. However, there is still redemption restrictions set by the banks or limited by the KYC/AML rules. Wechat Pay Balance is still operating and is one of the largest digital payment methods in China.
USDT (Tether)	USDT is a stablecoin issued by Tether in 2014. It provide a unit of account in cryptocurrency trading and on-chain payments, particularly without the need for bank or regulatory permission. USDT is backed by US Tresury (By Genius Act), gold, Bitcoin and other financial assets. These assets are under the custody of Banks, Asset Management Companies, and other onshore and offshore financial firms.	Issuers redeem the underlying assets and transfer money to clients. Retail token redemptions (on-ramp, off-ramp) rely heavily on OTC third-party service providers and peer-to-peer token-to-fiat exchanges. The redemption fee charged by Tether is 0.1%. The minumum amount of token redemption is $100,000. USDT and other fiat-backed stablecoins continue to grow after Circle's IPO and Genius Act.
DAI	DAI is a decentralized collateral-backed currency issued by MakerDAO in 2018. It is backed by Bitcoin, Ether, and other tokens. Issuers liquidate backed assets or implement other strategies to support the token value pegs to the US dollar or other fiat currencies. There is no custodian for the DAI. It is widely used in decentralized cryptocurrency exchanges and other decentralized finance protocols.	To retrieve a portion or all of the collateral, the user must pay down or completely pay back the Dai she generated, plus the Stability Fee (5%) that continuously accrues on the Dai outstanding. The Stability Fee can only be paid in Dai.

Notes: This table reports the major redemption based currency issuance in history.

Table B1. Redemption activity, full sample vs. analysis sample

Merchant Type	Redemption Volume (Percentage of Toal)	Redemption Transaction (Percentage of Toal)	Amount Per Transaction (CAD)	# Transactions Per Merchant
	Panel A: All users			
Cafes	19%	38%	7.92	46.2
Retail Shop	41%	21%	31.76	15.7
Bars	8%	13%	10.25	28.1
Restaurants	23%	25%	15.25	33.9
Service Shop	9%	4%	37.78	8.2
Total	1,134,767 CAD	70,439	16.11	26.4
	Panel B: Analysis sample			
Cafes	15%	32%	8.74	17.4
Retail Shop	47%	27%	33.49	9.0
Bars	5%	9%	11.04	9.0
Restaurants	23%	27%	16.50	16.8
Service Shop	10%	5%	39.51	4.4
Total	600,250 CAD	31,388	19.12	11.8

Notes: The table provides a comprehensive overview of the redemption patterns of all users and frequent users. Only users and redemption stores located in an area with a longitude between 79.11524° W and 79.63926° W and a latitude between 43.58100° N and 43.85546° N are included. Frequent users are defined as users with 20 item posts from April 2018 to August 2019.

Table B2. User activity, full sample vs. analysis sample

	Full Sample	Analysis Sample	Percentage
Number of users	193,989	7,162	3.69%
Total items posted (2018/04-2019/08)	1,044,234	834,194	79.89%
Total ratings received	772,328	466,050	60.34%
Total BTZ sent	390,282,261	221,650,823	56.79%
Total BTZ received	555,008,838	262,127,349	47.23%
Total BTZ holding	393,724,786	40,476,526	10.28%

Notes: The table provides a comprehensive comparison between the full sample users and analysis sample users. Full sample users are defined as users located in an area with a longitude between 79.11524° W and 79.63926° W and a latitude between 43.58100° N and 43.85546° N. Analysis sample users are defined as users with 20 item posts from April 2018 to August 2019.

Table B3. Effect of redemption exposure on token inflow, intensive vs. extensive margins

	(1) Asinh token inflow	(2) Asinh inflow transactions	(3) Any inflow transactions
Exposure	0.026***	0.005***	0.003***
	(0.005)	(0.001)	(0.001)
Baseline mean	2.046	0.350	0.250
# Obs	7,162	7,162	7,162

Notes: This table reports the effect of redemption exposure on token inflow. Columns (1) and (2) report the asinh ($asinh(x) = ln(x + \sqrt{x^2 + 1})$) amount of token inflow and number of transactions with token inflow, respectively. The analysis period is from April 2018 to August 2019. Robust standard deviations are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B4. Effect of redemption exposure on token outflow, intensive vs. extensive margins

	(1) Asinh token outflow	(2) Asinh outflow transactions	(3) Any outflow transactions
Exposure	0.002	0.001	0.0001
	(0.004)	(0.001)	(0.001)
Baseline mean	2.046	0.350	0.250
# Obs	7,162	7,162	7,162

Notes: This table reports the effect of redemption exposure on token outflow. Columns (1) and (2) report the asinh ($asinh(x) = ln(x + \sqrt{x^2 + 1})$) amount of token outflow and number of transactions with token outflow, respectively. The analysis period is from April 2018 to August 2019. Robust standard deviations are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B5. Effect of redemption exposure on token redemption, intensive vs. extensive margins

	(1) Asinh token redeemed	(2) Asinh redemption transactions	(3) Any redemption transaction
Exposure	0.029***	0.005***	0.003***
	(0.003)	(0.001)	(0.000)
Baseline mean	0.511	0.081	0.060
# Obs	7,162	7,162	7,162

Notes: This table reports the effect of redemption exposure on token redemption. Columns (1) and (2) report the asinh ($asinh(x) = ln(x + \sqrt{x^2 + 1})$) amount of token redeemed and number of redemption transactions, respectively. The analysis period is from April 2018 to August 2019. Robust standard deviations are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B6. Effect of redemption exposure on token issuance, intensive vs. extensive margins

	(1) Asinh token issued	(2) Asinh issuance transactions	(3) Any issuance transactions
Exposure	0.004 (0.004)	0.003 (0.003)	0.001 (0.001)
Baseline mean	4.770	2.724	0.745
# Obs	7,162	7,162	7,162

Notes: This table reports the effect of redemption exposure on token issuance. Columns (1) and (2) report the asinh ($asinh(x) = ln(x + \sqrt{x^2 + 1})$) amount of token issued and number of transactions with token inflow from Bunz, respectively. The analysis period is from April 2018 to August 2019. Robust standard deviations are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B7. Horse race: food v.s. retail merchant exposure

	(1) Acceptance	(2) Inflow	(3) Redemption	(4) Outflow	(5) Issuance	(6) Holding
# Nearby food	0.006***	0.072***	0.044***	0.033**	0.020***	0.018**
merchants	(0.001)	(0.013)	(0.016)	(0.014)	(0.006)	(0.007)
# Nearby retail	0.001	0.016	0.081***	-0.029***	-0.013***	-0.001
merchants	(0.001)	(0.011)	(0.012)	(0.011)	(0.005)	(0.006)
# Obs	7,162	7,162	7,162	7,162	7,162	7,162

Notes: Table shows the horse-race effect of food and retail redemption exposure on the token-related activity of users. The food and retail redemption exposure are defined as the number of food and retail merchants within 1 km of users, respectively. The analysis period is from April 2018 to August 2019.

Table B8. Horse race: food v.s. retail merchant exposure, with controls

	(1) Acceptance	(2) Inflow	(3) Redemption	(4) Outflow	(5) Issuance	(6) Holding
# Nearby food merchants	0.003	0.054***	0.007	0.032**	0.007***	0.018**
	(0.002)	(0.014)	(0.030)	(0.015)	(0.003)	(0.007)
# Nearby retail merchants	0.002	0.025*	0.099***	0.009	0.002	0.006
	(0.002)	(0.014)	(0.024)	(0.012)	(0.002)	(0.006)
Demographic controls	X	X	X	X	X	X
User activeness controls	X	X	X	X	X	X
Distance to city center	X	X	X	X	X	X
# Obs	2,204	2,204	2,204	2,204	2,204	2,204

Notes: Table shows the horse-race effect of food and retail redemption exposure on the token-related activity of users with additional controls. Specifically, this table further controls the demographic characteristics, including the age, income, and education of users, the activeness of users, including the number of new item posts and ratings sent to other users, and the distance between users' location and city center. The food and retail redemption exposure are defined as the number of food and retail merchants within 1 km of users, respectively. The analysis period is from April 2018 to August 2019.

Table B9. Effect of redemption exposure on token activity, restricted sample

	(1) Token acceptance	(2) Asinh token holdings	(3) Asinh token inflow	(4) Asinh token outflow	(5) Asinh token redeemed	(6) Asinh token issuance
# Nearby redemption merchant	0.003***	0.007	0.085***	0.002	0.046***	-0.001
	(0.001)	(0.007)	(0.013)	(0.005)	(0.011)	(0.011)
Baseline mean	0.250	7.801	5.159	5.697	2.216	6.717
# Obs	2,824	2,824	2,824	2,824	2,824	2,824

Notes: This table reports the impact of redemption exposure on token activity of the active users who also post at least 20 items before token introduction. The redemption exposure is defined as the number of merchants within 1 km of users. The analysis period is from April 2018 to August 2019. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B10. Effect of redemption exposure on user activeness

	(1) Asinh # item posts	(2) Asinh # item posts	(3) Asinh # ratings	(4) Asinh # ratings
Exposure	-0.003	0.005	-0.009*	-0.013
	(0.007)	(0.008)	(0.005)	(0.008)
Demographic	N	Y	N	Y
Distance	N	Y	N	Y
# Obs	7,162	2,204	7,162	2,204

Notes: Table shows the effect of redemption exposure on the activeness of users. The redemption exposure is defined as the number of merchants within 1 km of users. Columns (2) and (4) further controls the demographic characteristics, including the age, income, and education of users and the distance between users' location and city center. The analysis period is from April 2018 to August 2019.

Table B11. Effect of redemption exposure on token activity, alternative redemption exposure measurement

	Independent Variable					
Dependent Variable	# merchants within 1 km	asinh # merchants within 1 km	asinh average BTZ redeemed within 1 km	asinh average redemption transactions within 1 km	asinh average distance from merchants	asinh average distance from merchants weighted by redemption volume
Token acceptance	0.003***	0.015***	0.002***	0.005***	-0.030***	-0.031***
	(0.001)	(0.003)	(0.001)	(0.001)	(0.006)	(0.006)
R^2	0.005	0.004	0.002	0.002	0.003	0.003
Asinh token inflow	0.026***	0.094***	0.015***	0.034***	-0.153***	-0.167***
	(0.005)	(0.019)	(0.004)	(0.008)	(0.043)	(0.044)
R^2	0.005	0.003	0.002	0.002	0.002	0.002
Asinh token redeemed	0.026***	0.128***	0.023***	0.047***	-0.243***	-0.241***
	(0.003)	(0.011)	(0.002)	(0.004)	(0.024)	(0.025)
R^2	0.014	0.018	0.013	0.013	0.012	0.011
Asinh token holdings	0.003	-0.006	-0.007*	-0.011	0.114**	0.107**
	(0.004)	(0.019)	(0.004)	(0.008)	(0.046)	(0.047)
R^2	0.000	0.000	0.000	0.000	0.001	0.001
Asinh token issued	0.003	-0.010	-0.006*	-0.010	0.098***	0.090**
	(0.004)	(0.015)	(0.003)	(0.007)	(0.036)	(0.037)
R^2	0.000	-0.000	0.000	0.000	0.001	0.001
Asinh token outflow	0.002	-0.012	0.000	0.002	0.026	0.011
	(0.004)	(0.018)	(0.004)	(0.008)	(0.043)	(0.044)
R^2	0.000	0.000	0.000	0.000	0.000	0.000

Notes: This table reports the effect of redemption exposure on token related behavior using alternative redemption exposure measurement. The analysis period is from April 2018 to August 2019. Robust standard deviations are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B12. Effect of redemption exposure on token inflows and outflows, alternative scales

	Panel A: Asinh token inflow			
	(1) y	(2) 0.01 × y	(3) 100 × y	(4) 10,000 × y
# Nearby redemption merchant	0.026*** (0.005)	0.013*** (0.002)	0.039*** (0.007)	0.052*** (0.009)
	Panel B: Asinh token outflow			
# Nearby redemption merchant	0.002 (0.004)	0.001 (0.002)	0.002 (0.007)	0.002 (0.009)
# Obs	7,162	7,162	7,162	7,162

Notes: Table shows the effect of redemption exposure on asinh amount of token inflow and outflow at different scales. Specifically, the dependent variables are scaled to 1% (Column 2), 100 times (Column 3), and 10,000 times (Column 4) the original flow volume. The redemption exposure is defined as the number of merchants within 1 km of users. The analysis period is from April 2018 to August 2019.

Table B13. Effect of redemption exposure on token redemption and issuance, alternative scales

	Panel A: Asinh token redemption			
	(1)	(2)	(3)	(4)
	y	0.01 × y	100 × y	10,000 × y
# Nearby redemption merchant	0.029***	0.013***	0.045***	0.061***
	(0.003)	(0.002)	(0.005)	(0.006)
	Panel B: Asinh token issuance			
# Nearby redemption merchant	0.004	0.001	0.006	0.009
	(0.004)	(0.002)	(0.006)	(0.009)
# Obs	7,162	7,162	7,162	7,162

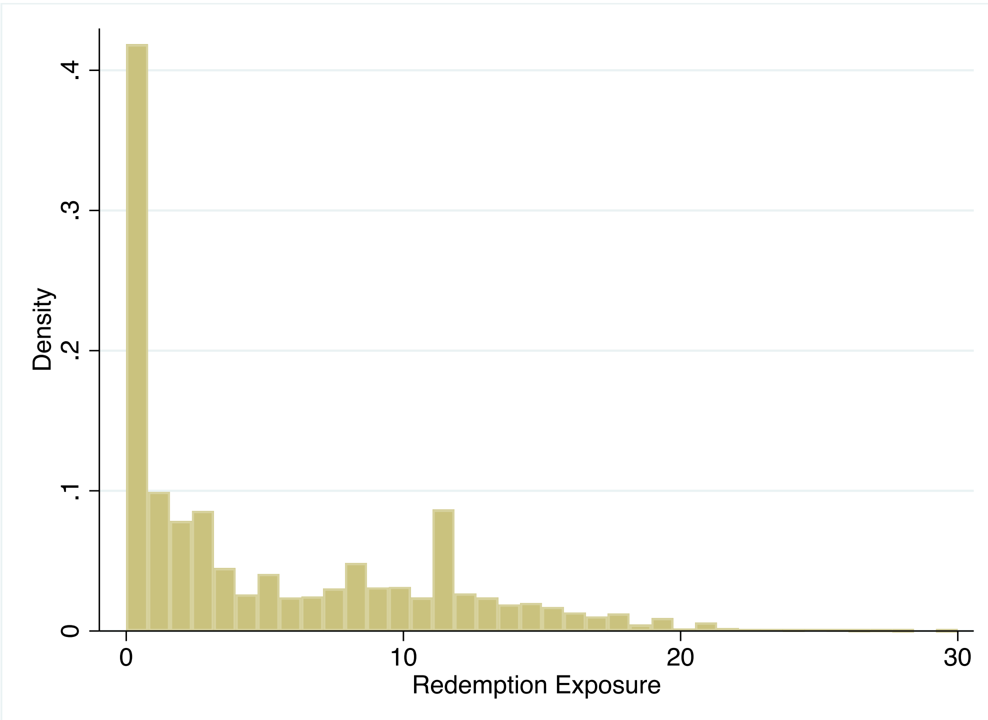
Notes: Table shows the effect of redemption exposure on asinh amount of token redemption and issuance at different scales. Specifically, the dependent variables are scaled to 1% (Column 2), 100 times (Column 3), and 10,000 times (Column 4) the original flow volume. The redemption exposure is defined as the number of merchants within 1 km of users. The analysis period is from April 2018 to August 2019.

Table B14. Effect of redemption exposure on token flows, Poisson regression

	(1) Token inflow	(2) Token outflow	(3) Token redemption	(4) Token issuance
# Nearby redemption merchant	0.020***	0.011**	0.036***	-0.001
	(0.005)	(0.005)	(0.005)	(0.002)
# Obs	7,162	7,162	7,162	7,162

Notes: Table shows the effect of redemption exposure on asinh amount of token flows using Poisson regression. The redemption exposure is defined as the number of merchants within 1 km of users. The analysis period is from April 2018 to August 2019.

Figure C1. Distribution of redemption exposure

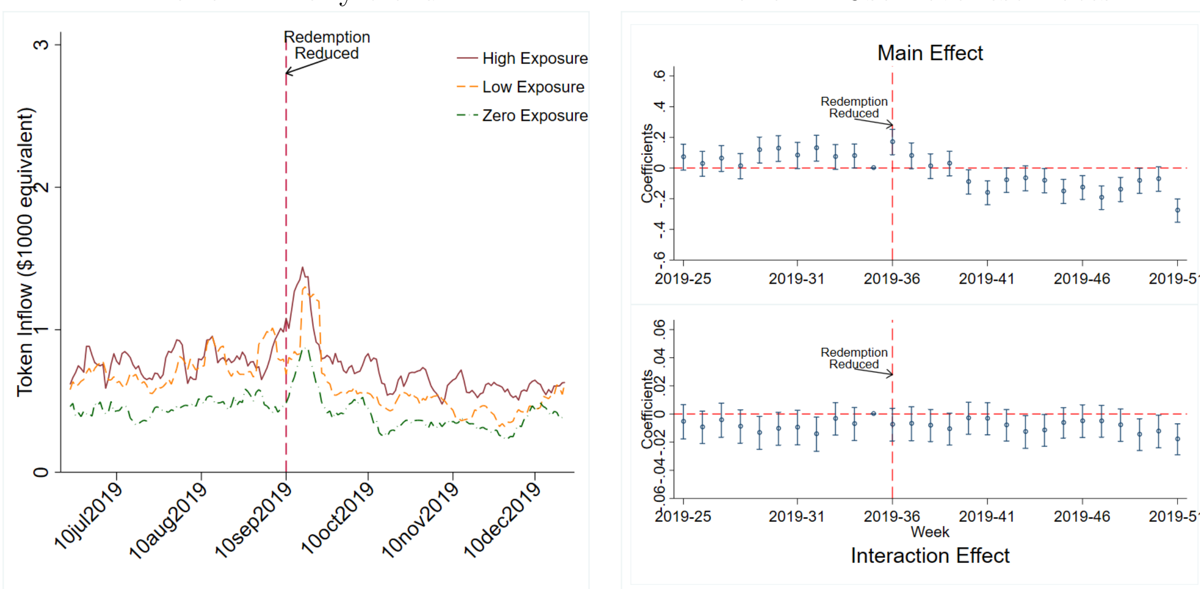


Notes: The figure plots the distribution of the average number of merchants within 1 km of active users from April 2018 to August 2019.

Figure C2. Effects of reduced redeemability on token inflow

Panel A: Daily trend

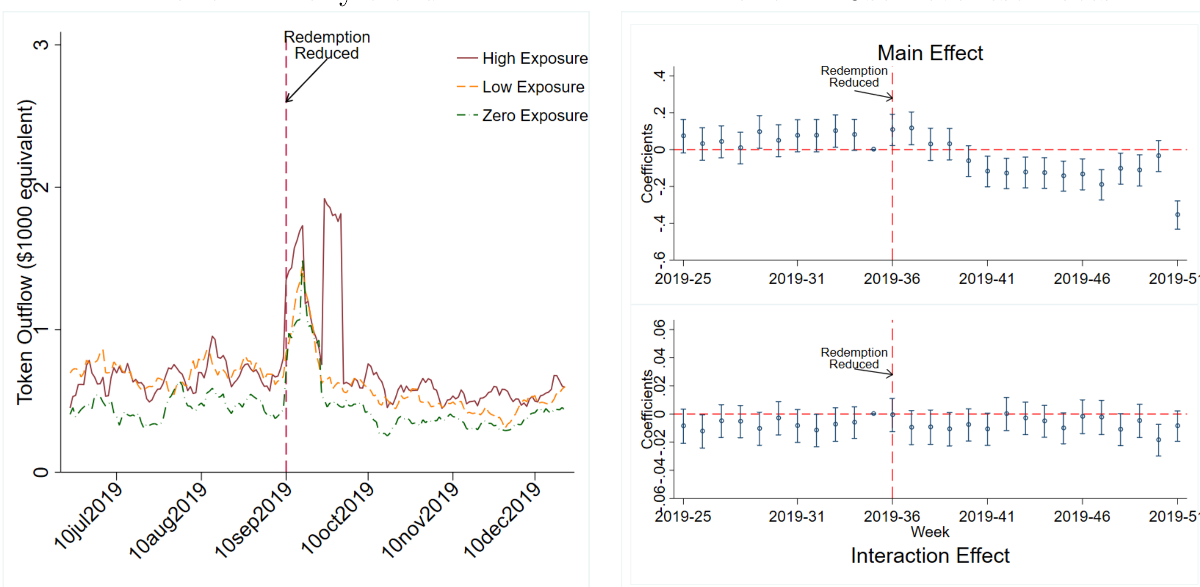
Panel B: User-level estimates



Notes: The figure plots the effect of reduced redeemability on token inflow. Panel A shows the 7-day moving average amount of token inflow by redemption exposure. Panel B plot the regression coefficients $\beta_{I,k}$ and $\beta_{D,k}$, with the 95 percent confidence intervals, estimated by Equation (7) using the token inflow as outcome variable. The red dashed line indicates September 10, the day of partial cessation of Shop Local program.

Figure C3. Effects of reduced redeemability on token outflow

Panel A: Daily trend | Panel B: User-level estimates

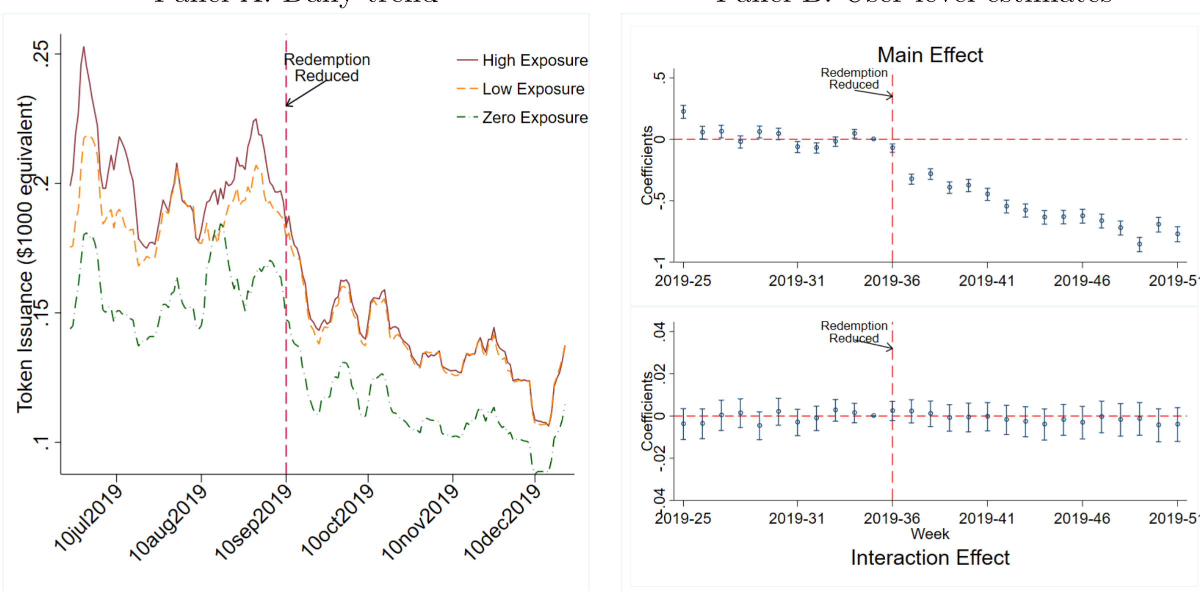


Notes: The figure plots the effect of reduced redeemability on token outflow. Panel A shows the 7-day moving average amount of token outflow by redemption exposure. Panel B plot the regression coefficients $\beta_{I,k}$ and $\beta_{D,k}$, with the 95 percent confidence intervals, estimated by Equation (7) using the token outflow as outcome variable. The red dashed line indicates September 10, the day of partial cessation of Shop Local program.

Figure C4. Effects of reduced redeemability on token issuance

Panel A: Daily trend

Panel B: User-level estimates

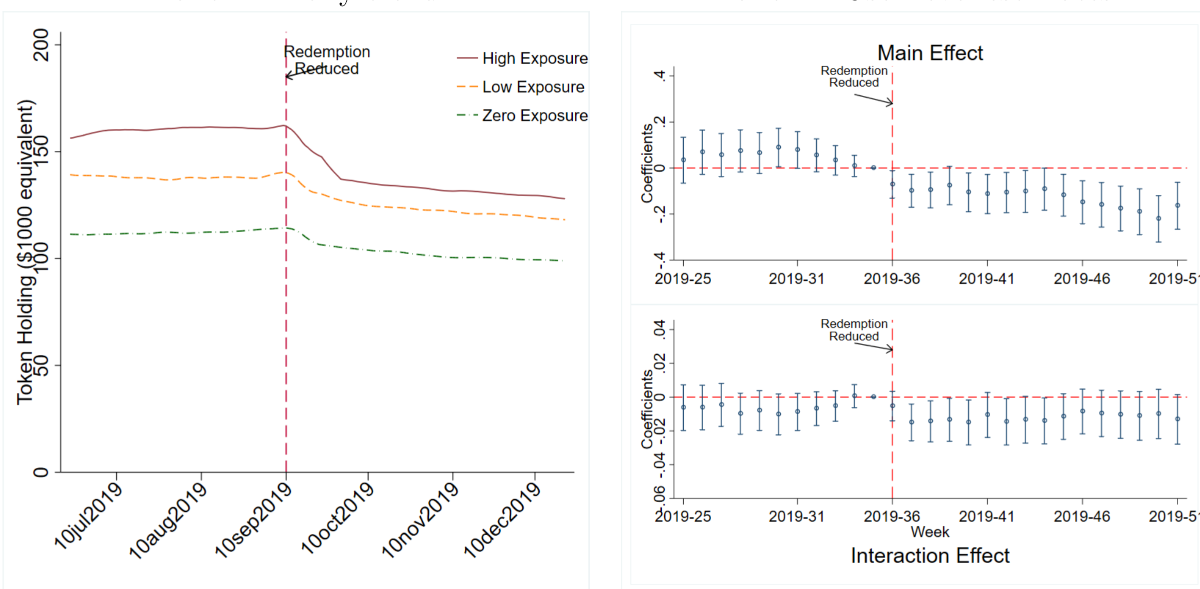


Notes: The figure plots the effect of reduced redeemability on token issuance. Panel A shows the 7-day moving average amount of token issuance by redemption exposure. Panel B plot the regression coefficients $\beta_{I,k}$ and $\beta_{D,k}$, with the 95 percent confidence intervals, estimated by Equation (7) using the token issuance as outcome variable. The red dashed line indicates September 10, the day of partial cessation of Shop Local program.

Figure C5. Effects of reduced redeemability on token holdings

Panel A: Daily trend

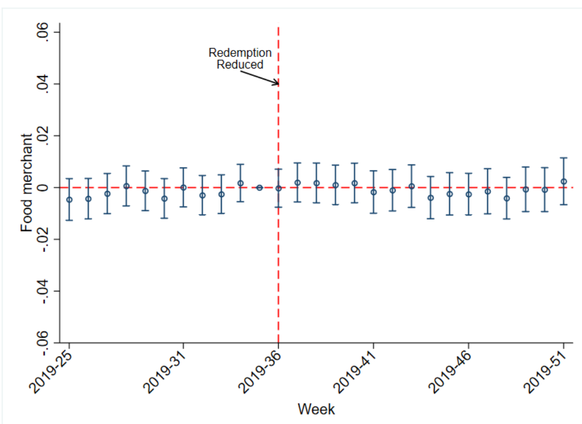
Panel B: User-level estimates



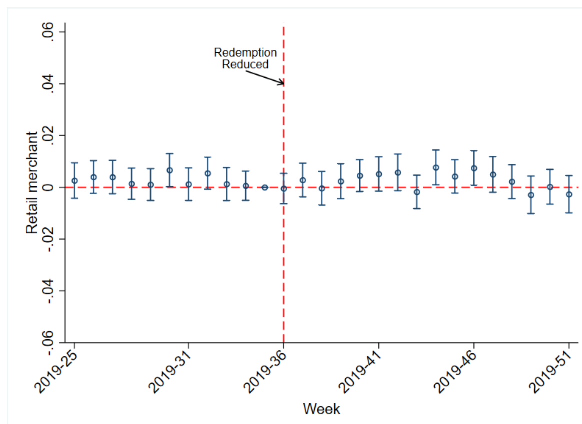
Notes: The figure plots the effect of reduced redeemability on token holdings. Panel A shows the 7-day moving average amount of token holdings by redemption exposure. Panel B plot the regression coefficients $\beta_{I,k}$ and $\beta_{D,k}$, with the 95 percent confidence intervals, estimated by Equation (7) using the token holdings as outcome variable. The red dashed line indicates September 10, the day of partial cessation of Shop Local program.

Figure C6. Redemption exposure and token acceptance under reduced redeemability: food vs. retail

Panel A: Food merchant



Panel B: Retail merchant

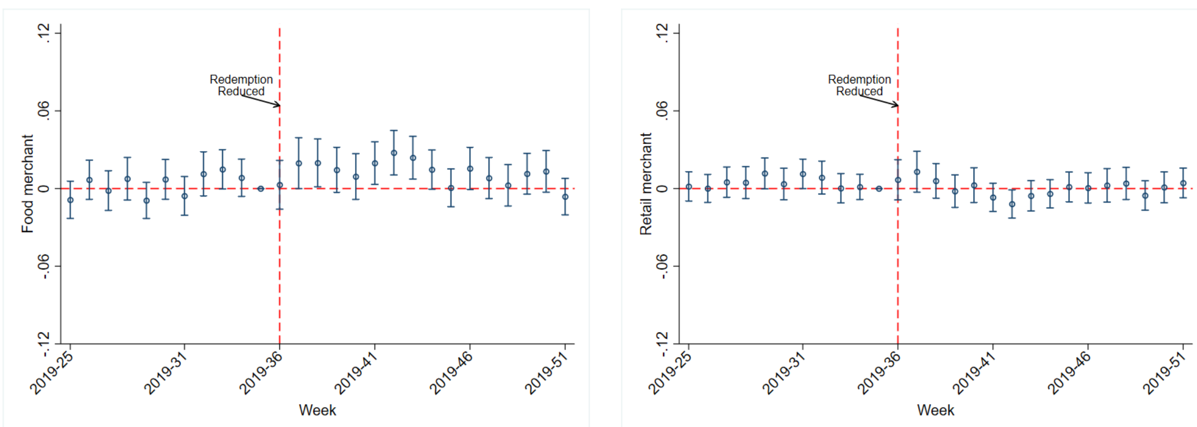


Notes: This figure plots the differential response of the effect of redemption exposure on token acceptance to reduced redeemability, comparing food merchants with retail merchants. Panels A and B display the estimated regression coefficients ($\beta_{f,k}$ for food and $\beta_{r,k}$ for retail, respectively) along with their 95% confidence intervals. These coefficients were estimated using Equation (8), with token acceptance as the outcome variable. The vertical dashed line at September 10 marks the partial cessation of the Shop Local program.

Figure C7. Redemption exposure and token redemption under reduced redeemability: food vs. retail

Panel A: Food merchant

Panel B: Retail merchant

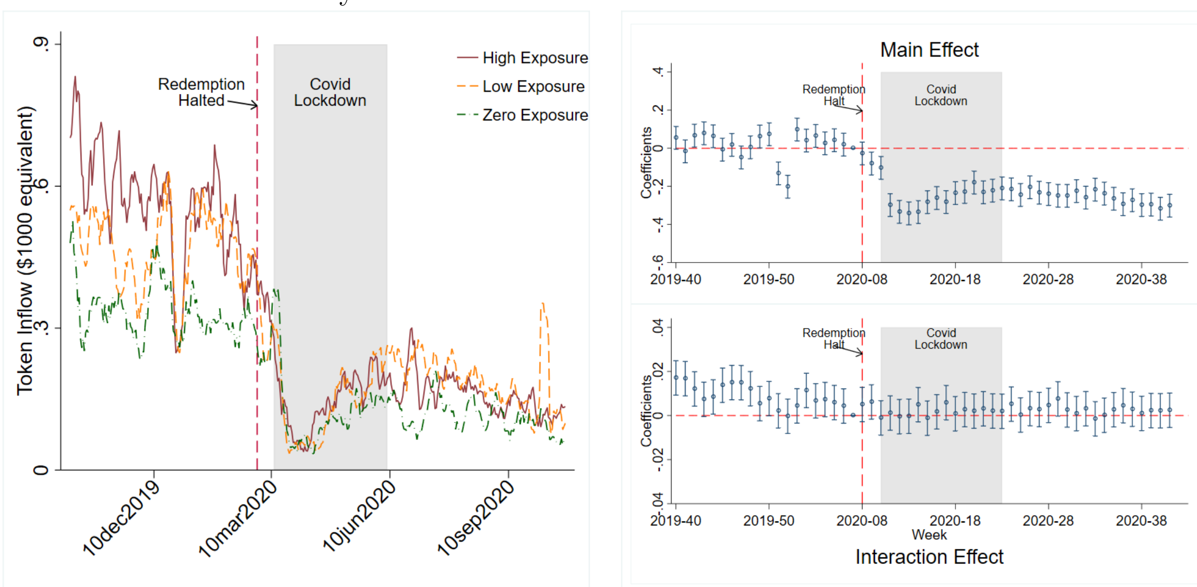


Notes: This figure plots the differential response of the effect of redemption exposure on token redemption to reduced redeemability, comparing food merchants with retail merchants. Panels A and B display the estimated regression coefficients ($\beta_{f,k}$ for food and $\beta_{r,k}$ for retail, respectively) along with their 95% confidence intervals. These coefficients were estimated using Equation (8), with token redemption as the outcome variable. The vertical dashed line at September 10 marks the partial cessation of the Shop Local program.

Figure D1. Effects of redemption halt on token inflow

Panel A: Daily trend

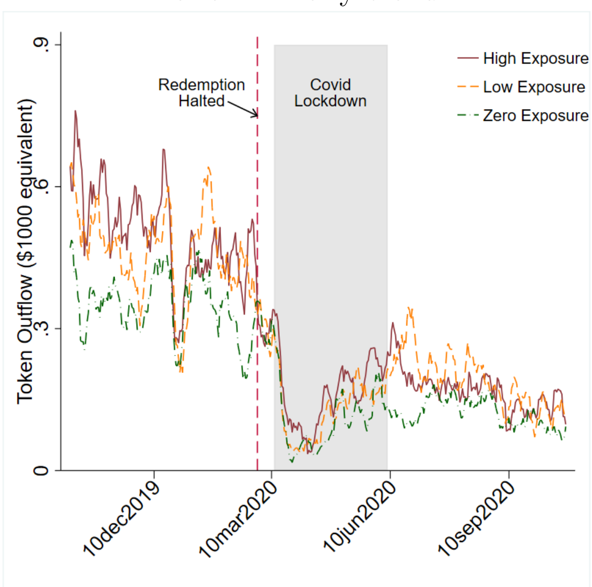
Panel B: User-level estimates



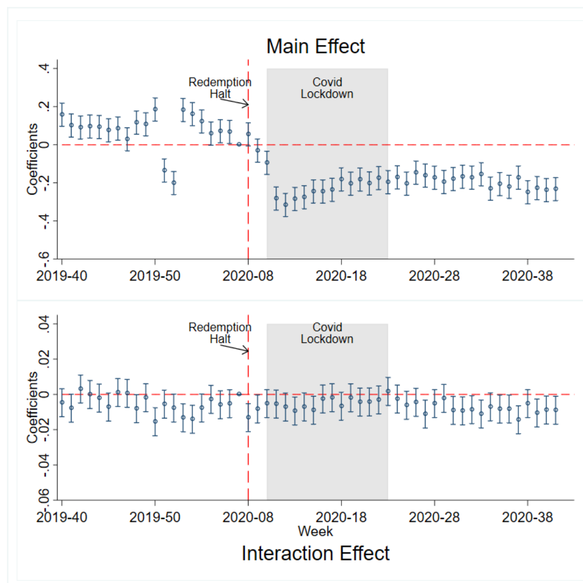
Notes: The figure plots the effect of redemption halt on token inflow. Panel A shows the 7-day moving average amount of token inflow by redemption exposure. Panel B plot the regression coefficients $\beta_{I,k}$ and $\beta_{D,k}$, with the 95 percent confidence intervals, estimated by Equation (9) using the token inflow as outcome variable. The red dashed line indicates February 28, the day of full cessation of Shop Local program.

Figure D2. Effects of redemption halt on token outflow

Panel A: Daily trend



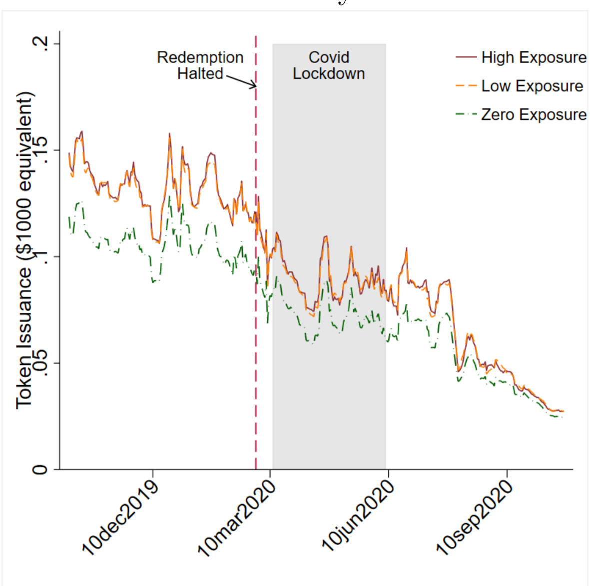
Panel B: User-level estimates



Notes: The figure plots the effect of redemption halt on token outflow. Panel A shows the 7-day moving average amount of token outflow by redemption exposure. Panel B plot the regression coefficients $\beta_{I,k}$ and $\beta_{D,k}$, with the 95 percent confidence intervals, estimated by Equation (9) using the token outflow as outcome variable. The red dashed line indicates February 28, the day of full cessation of Shop Local program.

Figure D3. Effects of redemption halt on token issuance

Panel A: Daily trend

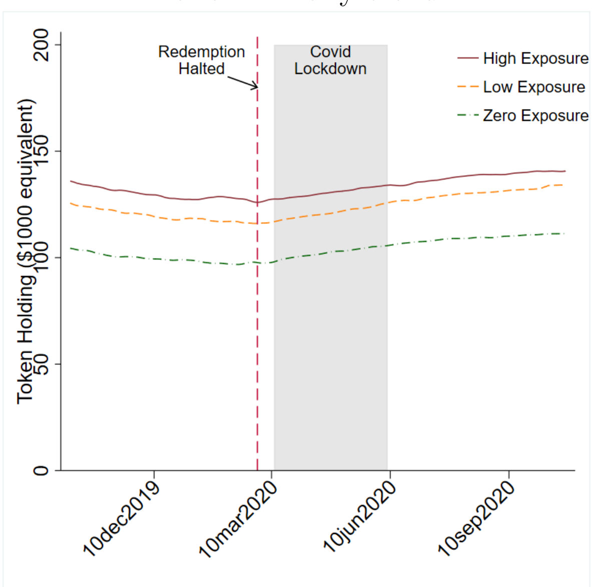


Panel B: User-level estimates

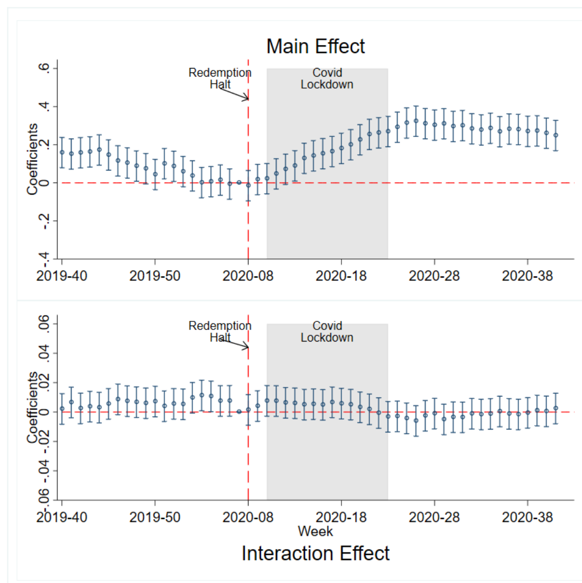
Notes: The figure plots the effect of redemption halt on token issuance. Panel A shows the 7-day moving average amount of token issued by redemption exposure. Panel B plot the regression coefficients $\beta_{I,k}$ and $\beta_{D,k}$, with the 95 percent confidence intervals, estimated by Equation (9) using the token issuance as outcome variable. The red dashed line indicates February 28, the day of full cessation of Shop Local program.

Figure D4. Effects of redemption halt on token holdings

Panel A: Daily trend



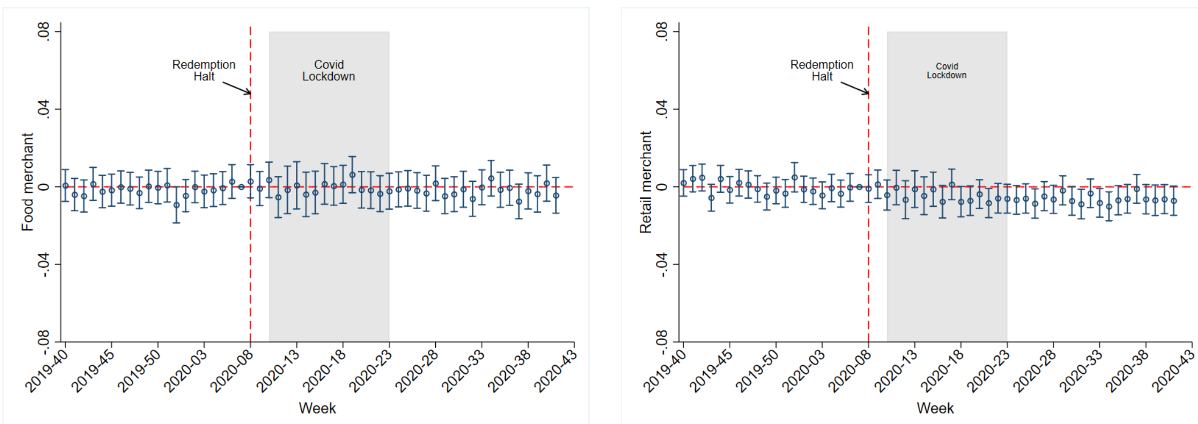
Panel B: User-level estimates



Notes: The figure plots the effect of redemption halt on token holdings. Panel A shows the 7-day moving average amount of token holding by redemption exposure. Panel B plot the regression coefficients $\beta_{I,k}$ and $\beta_{D,k}$, with the 95 percent confidence intervals, estimated by Equation (9) using the token holding as outcome variable. The red dashed line indicates February 28, the day of full cessation of Shop Local program.

Figure D5. redemption exposure and token acceptance under redemption halt: food vs. retail

Panel A: Food merchant | Panel B: Retail merchant



Notes: This figure plots the differential response of the effect of redemption exposure on token acceptance to redemption halt, comparing food merchants with retail merchants. Panels A and B display the estimated regression coefficients ($\beta_{f,k}$ for food and $\beta_{r,k}$ for retail, respectively) along with their 95% confidence intervals. These coefficients were estimated using Equation (10), with token acceptance as the outcome variable. The vertical dashed line at September 10 marks the partial cessation of the Shop Local program.